Trees

Walking a Tree

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Often we want to visit the nodes of a tree in a particular order.

For example, print the nodes of the tree.

- Depth-first: We completely traverse one sub-tree before exploring a sibling sub-tree.
- Breadth-first: We traverse all nodes at one level before progressing to the next level.

Depth-first

In-order traversal is to traverse the left subtree first. Then visit the root. Finally, traverse the right subtree.

```
def inorder(self):
    if self.left is not None:
        self.left.inorder()
    print(self.val)
    if self.right is not None:
        self.right.inorder()
```

InOrderTraversal

```python
def inorder(self):
    if self.left is not None:
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    print(self.val)
    if self.right is not None:
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```

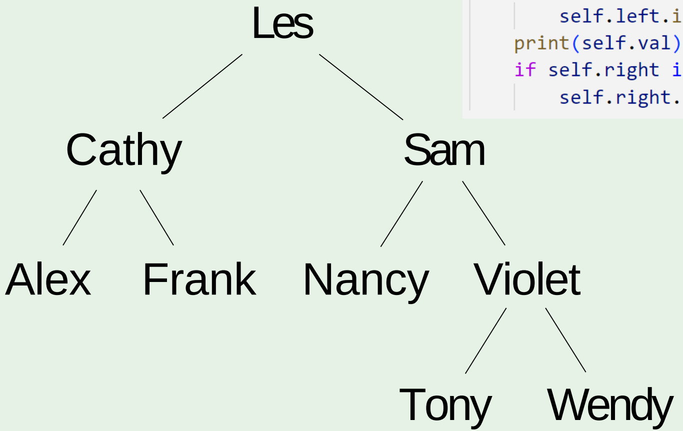

Output:

InOrderTraversal

```python
def inorder(self):
    if self.left is not None:
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    print(self.val)
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```

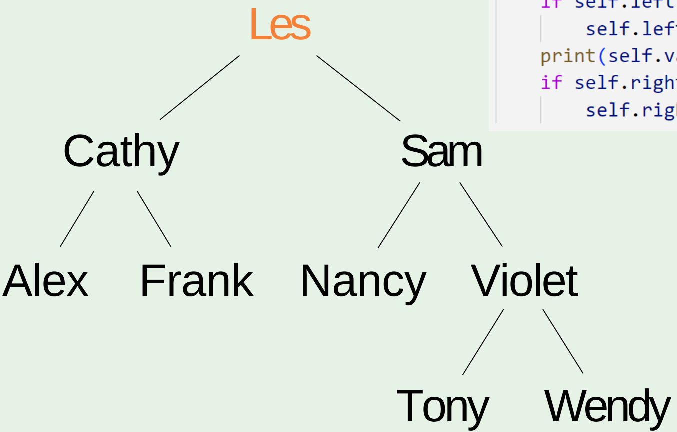

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InOrderTraversal

```python
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    if self.left is not None:
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    print(self.val)
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```

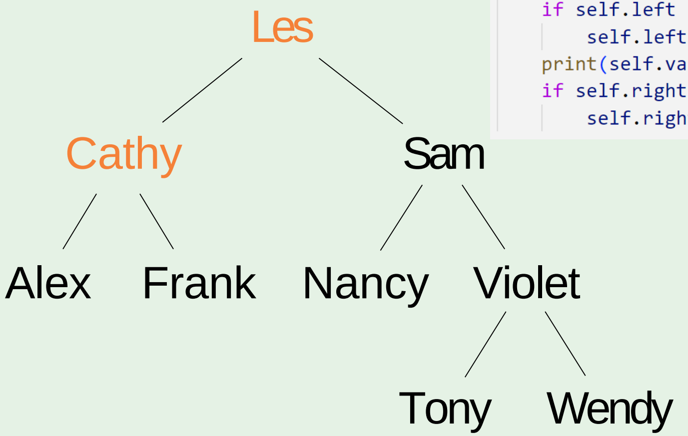

Output:

InOrderTraversal

```python
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    print(self.val)
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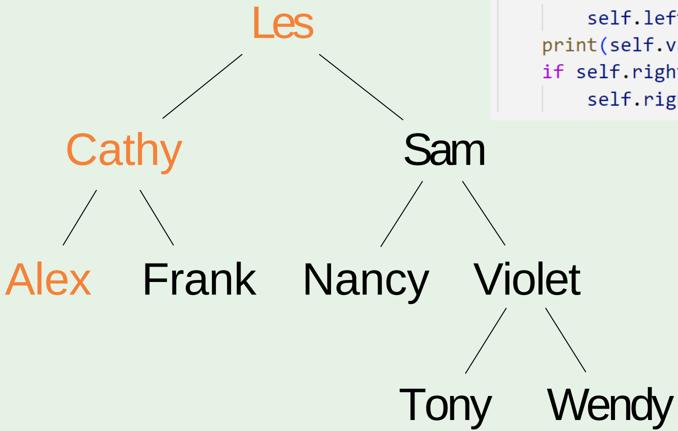

Output:

InOrderTraversal

```
def inorder(self):
    if self.left is not None:
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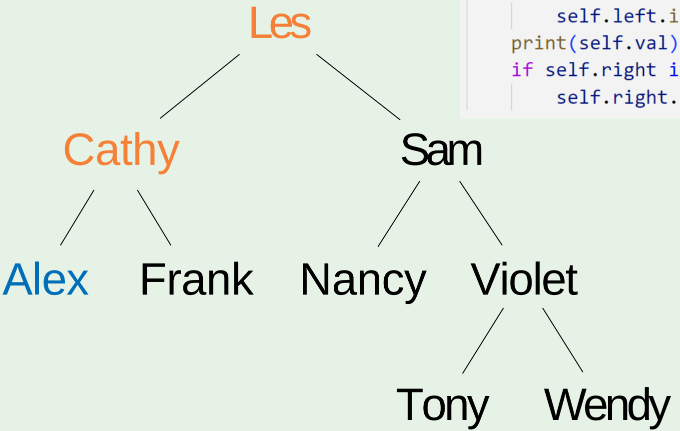

Output: Alex

InOrderTraversal

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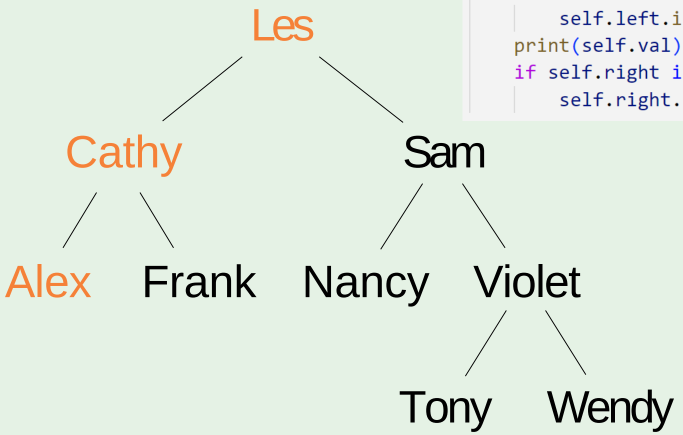

Output: Alex

InOrderTraversal

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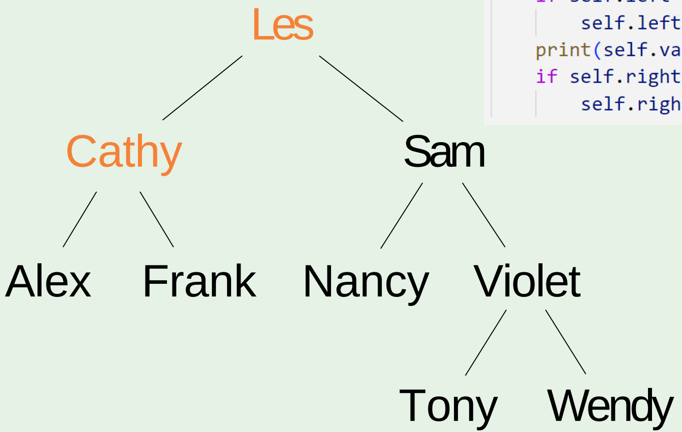

Output: Alex

InOrderTraversal

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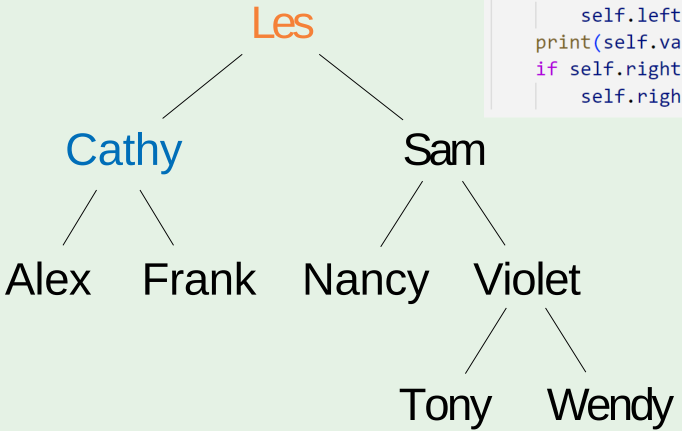

Output: Alex Cathy

InOrderTraversal

```
def inorder(self):
    if self.left is not None:
        self.left.inorder()
    print(self.val)
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```

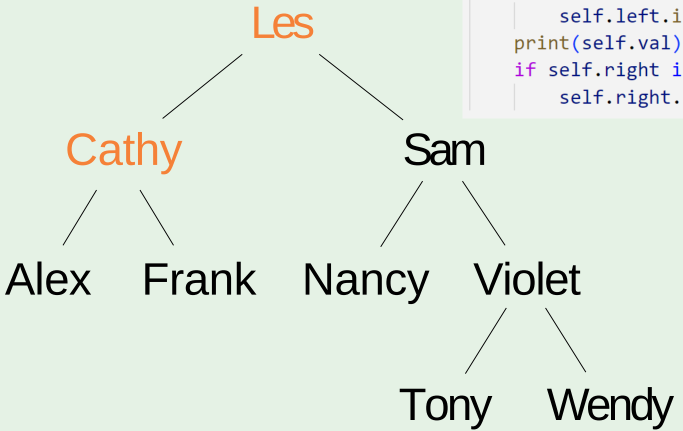

Output: Alex Cathy

InOrderTraversal

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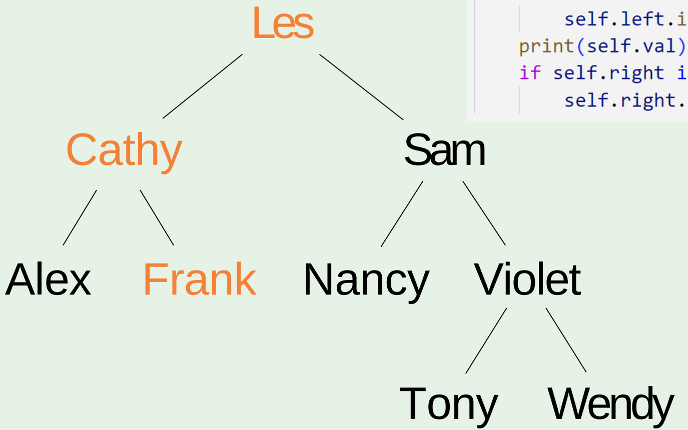

Output: Alex Cathy

InOrderTraversal

```
def inorder(self):
    if self.left is not None:
        self.left.inorder()
    print(self.val)
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```

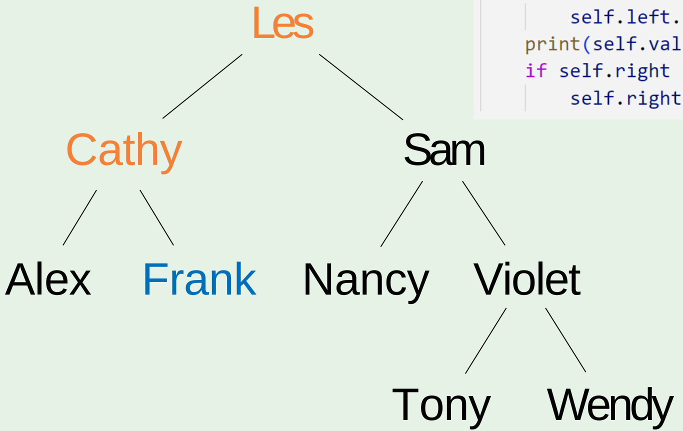

Output: Alex Cathy Frank

InOrderTraversal

```python
def inorder(self):
    if self.left is not None:
        self.left.inorder()
    print(self.val)
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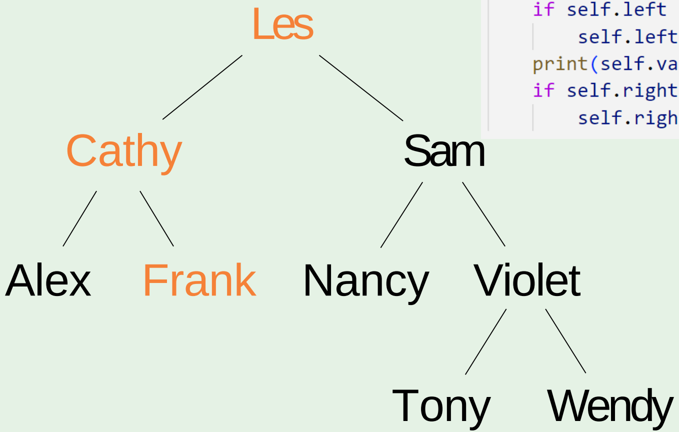

Output: Alex Cathy Frank

InOrderTraversal

```
def inorder(self):
    if self.left is not None:
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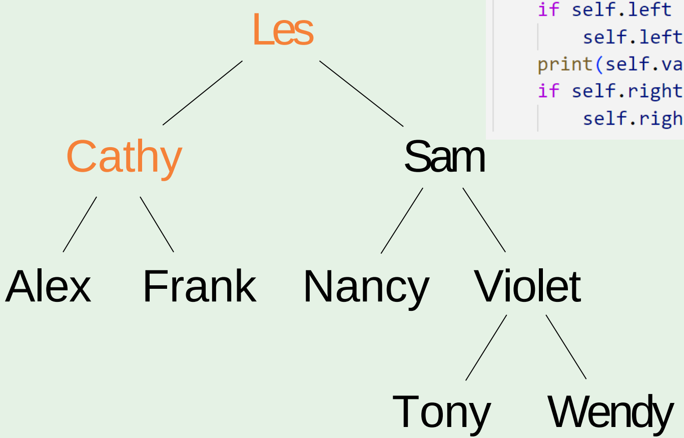

Output: Alex Cathy Frank

InOrderTraversal

```python
def inorder(self):
    if self.left is not None:
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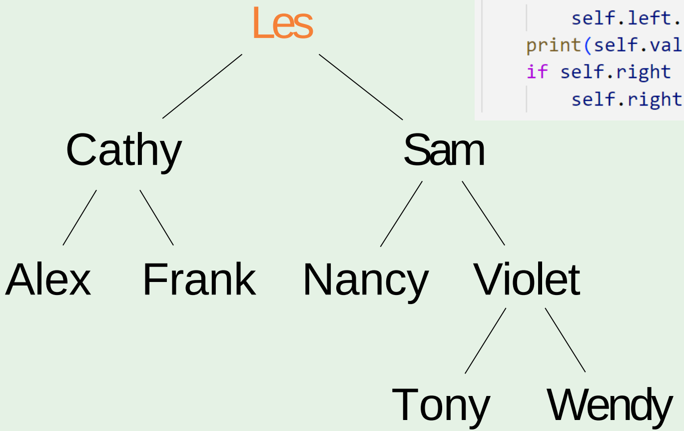

Output: Alex Cathy Frank

InOrderTraversal

```
def inorder(self):
    if self.left is not None:
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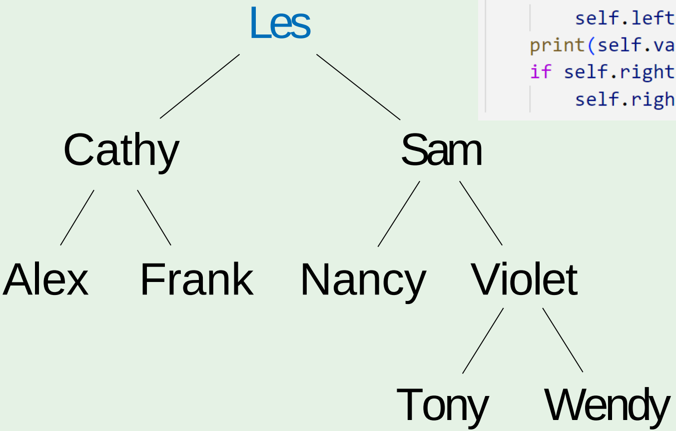

Output: Alex Cathy Frank Les

InOrderTraversal

```python
def inorder(self):
    if self.left is not None:
        self.left.inorder()
    print(self.val)
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```

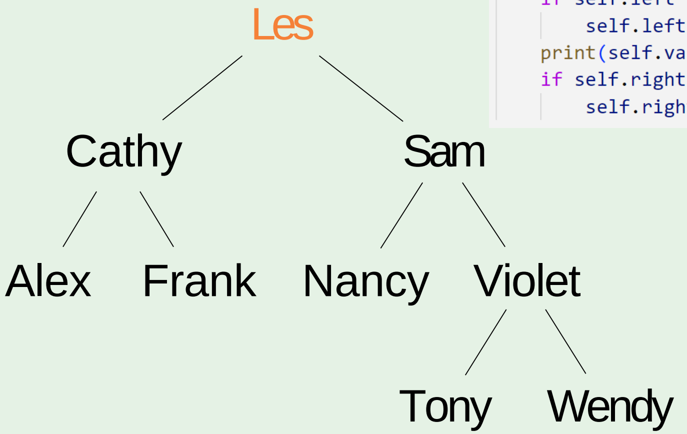

Output: Alex Cathy Frank Les

InOrderTraversal

```python
def inorder(self):
    if self.left is not None:
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    print(self.val)
    if self.right is not None:
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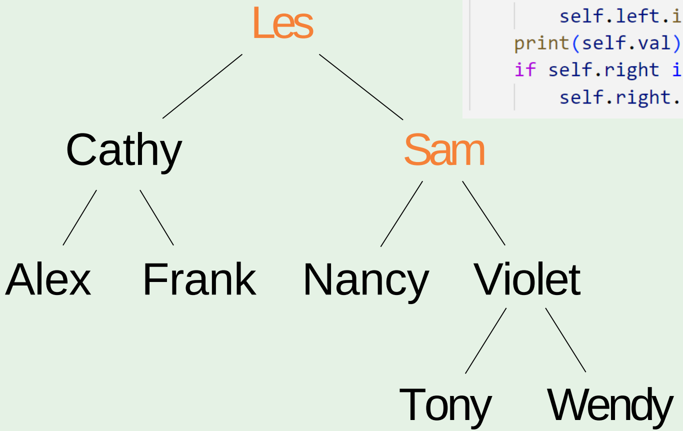

Output: Alex Cathy Frank Les

InOrderTraversal

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def inorder(self):
    if self.left is not None:
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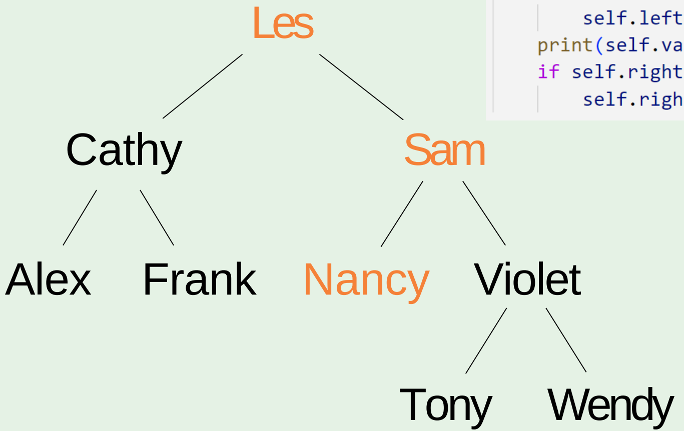

Output: Alex Cathy Frank Les

InOrderTraversal

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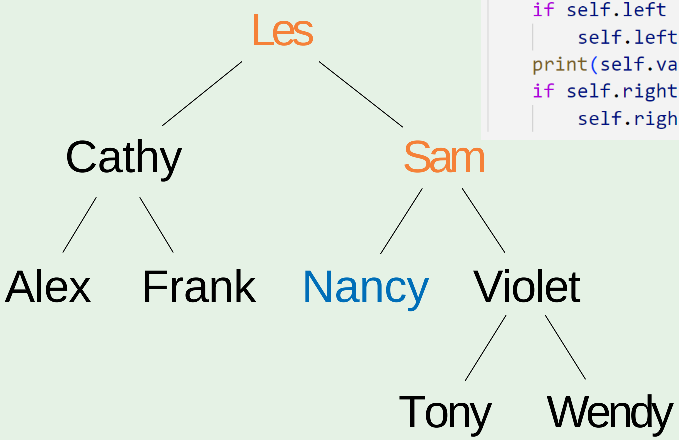

Output: Alex Cathy Frank Les Nancy

InOrderTraversal

```python
def inorder(self):
    if self.left is not None:
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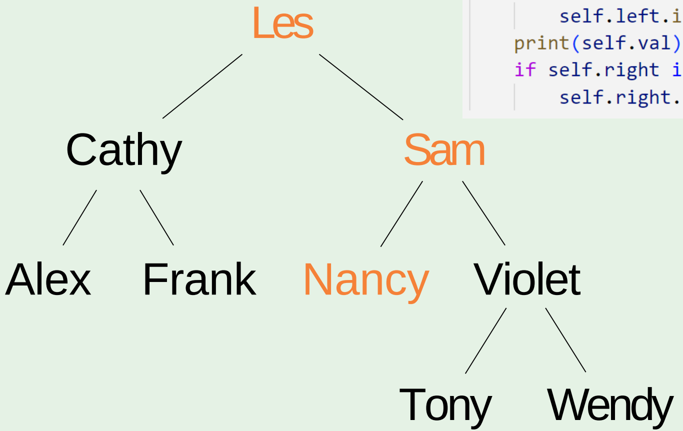

Output: Alex Cathy Frank Les Nancy

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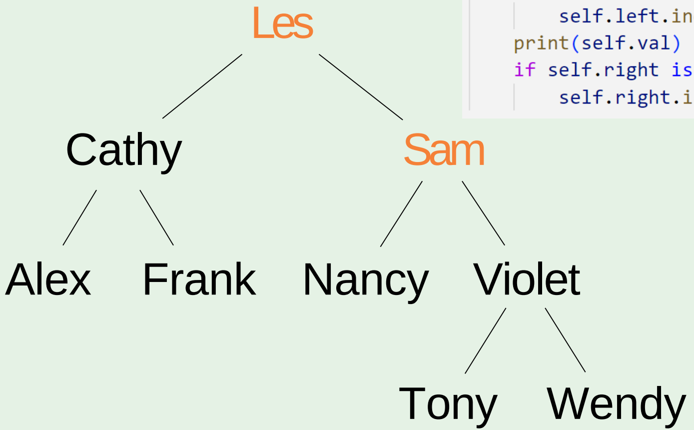

Output: Alex Cathy Frank Les Nancy

InOrderTraversal

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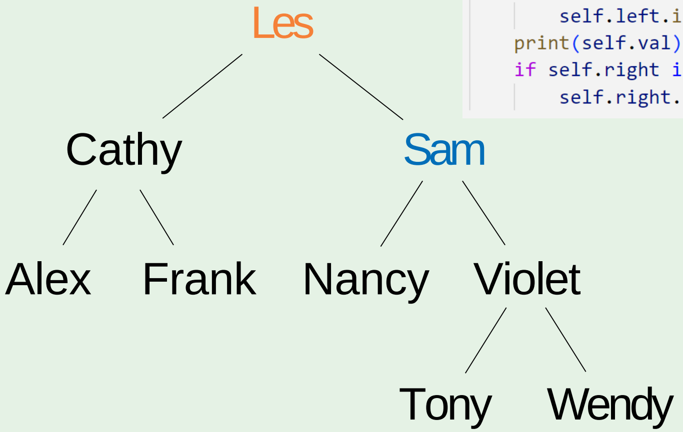

Output: Alex Cathy Frank Les Nancy Sam

InOrderTraversal

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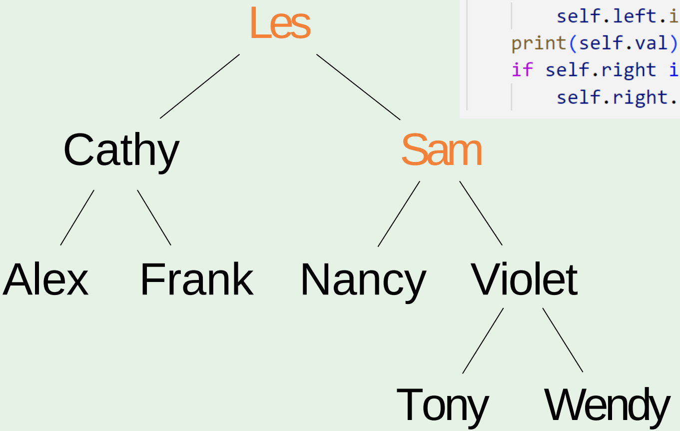

Output: Alex Cathy Frank Les Nancy Sam

InOrderTraversal

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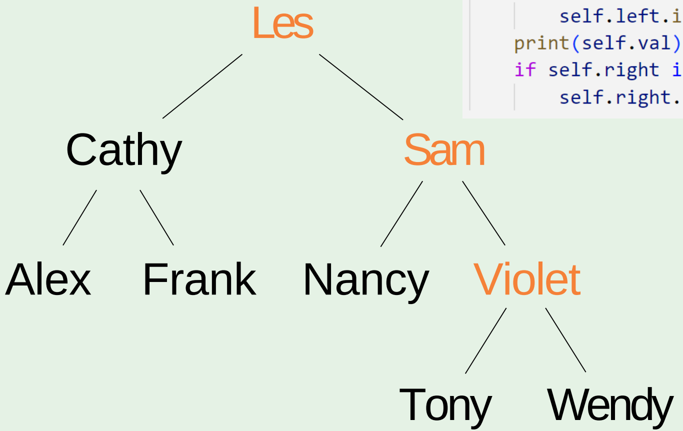

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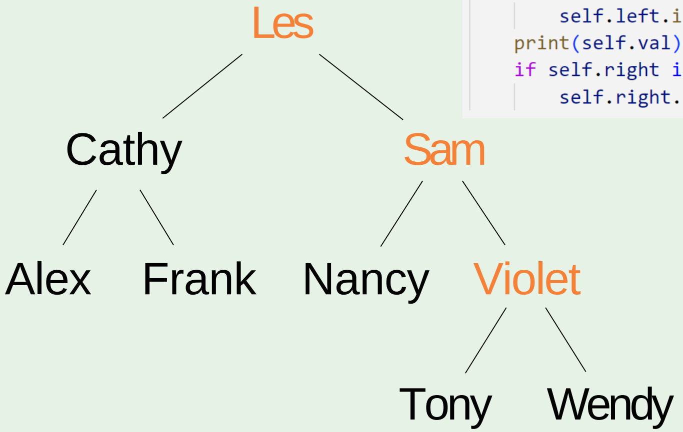

Output: Alex Cathy Frank Les Nancy Sam

InOrderTraversal

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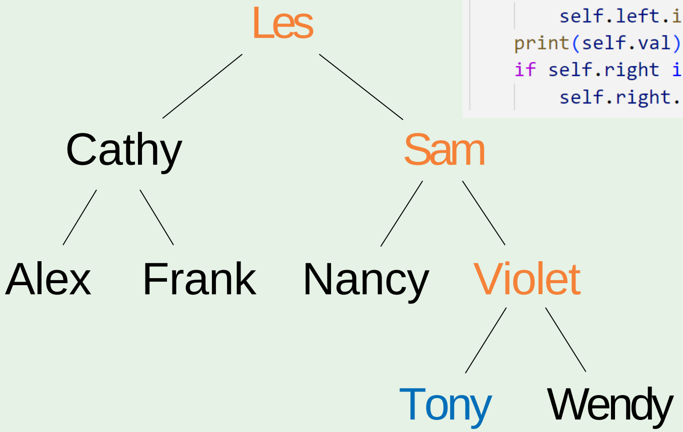

Output: Alex Cathy Frank Les Nancy Sam Tony

InOrderTraversal

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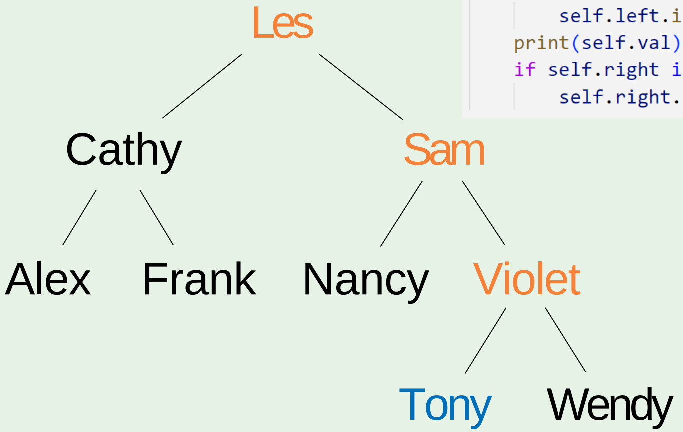

Output: Alex Cathy Frank Les Nancy Sam Tony

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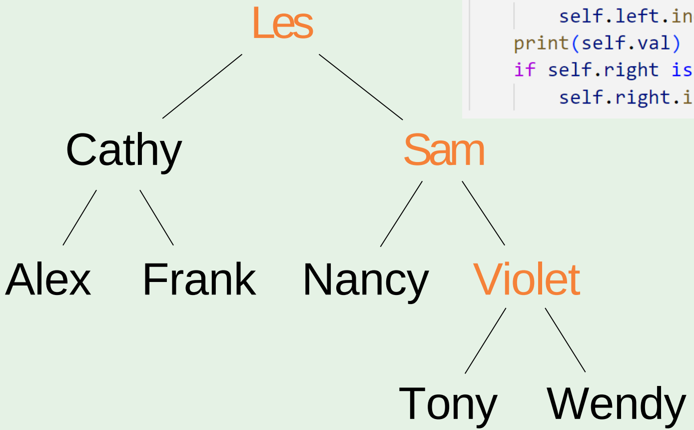

Output: Alex Cathy Frank Les Nancy Sam Tony

InOrderTraversal

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```

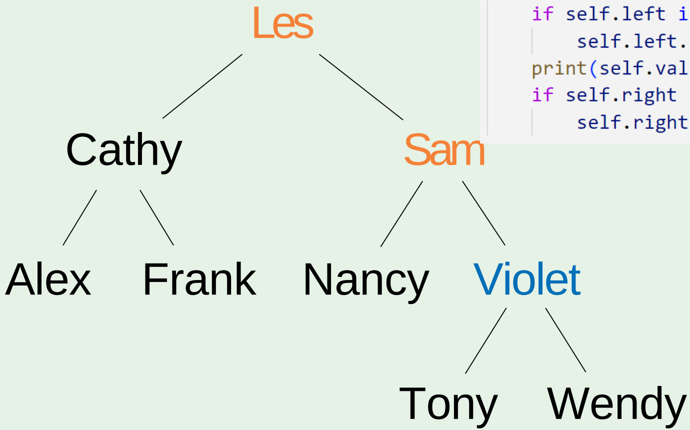

Output: Alex Cathy Frank Les Nancy Sam Tony Violet

InOrderTraversal

```python
def inorder(self):
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```

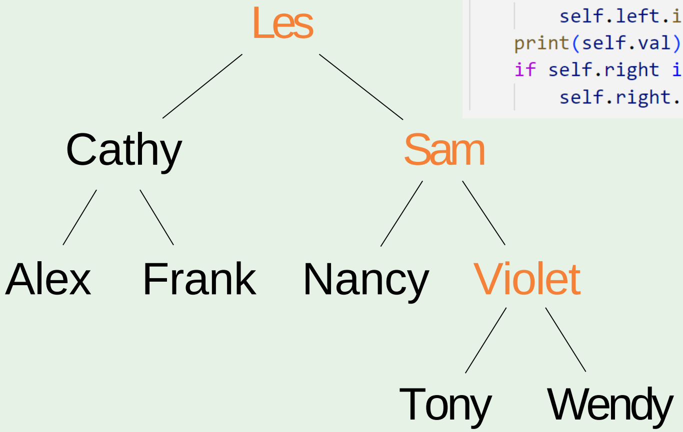

Output: Alex Cathy Frank Les Nancy Sam Tony Violet

InOrderTraversal

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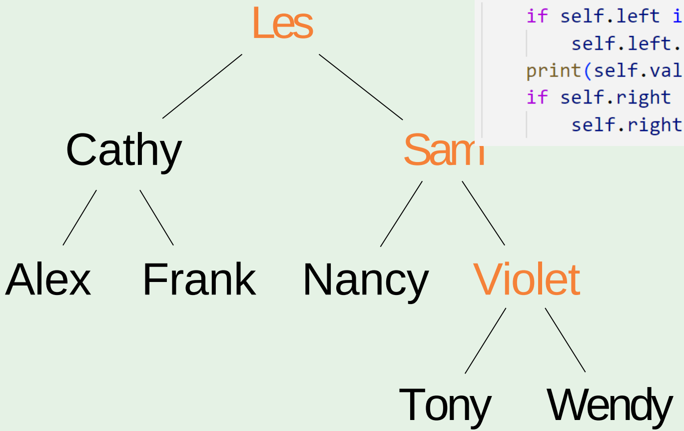

Output: Alex Cathy Frank Les Nancy Sam Tony Violet

InOrderTraversal

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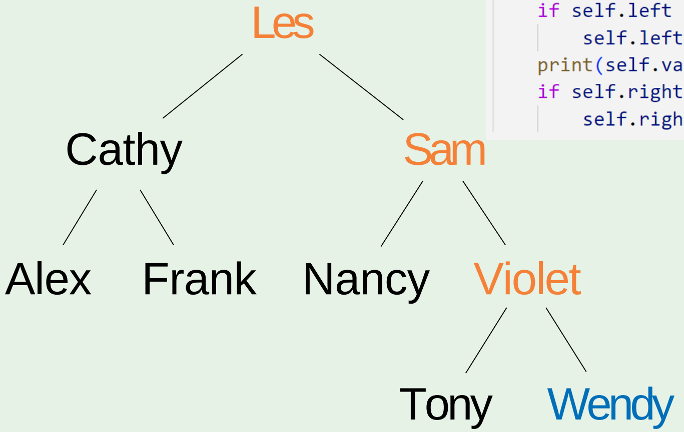

Output: Alex Cathy Frank Les Nancy Sam Tony Violet Wendy

InOrderTraversal

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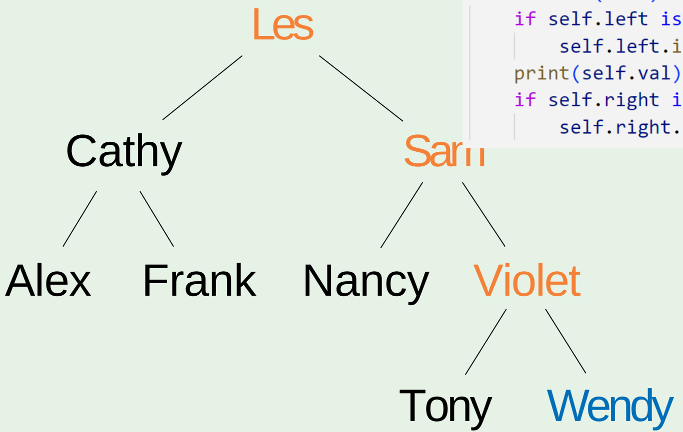

Output: Alex Cathy Frank Les Nancy Sam Tony Violet Wendy

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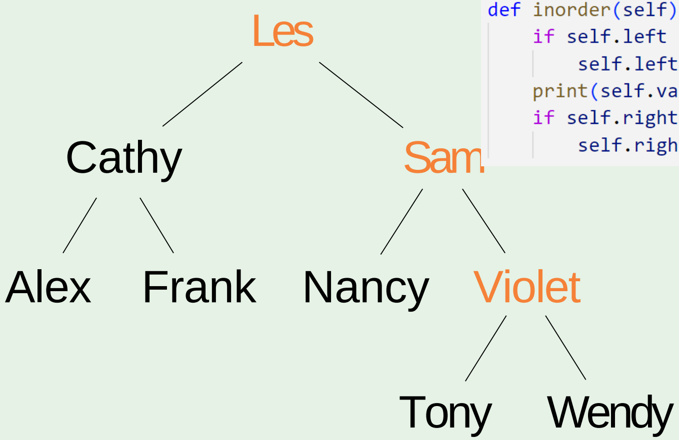
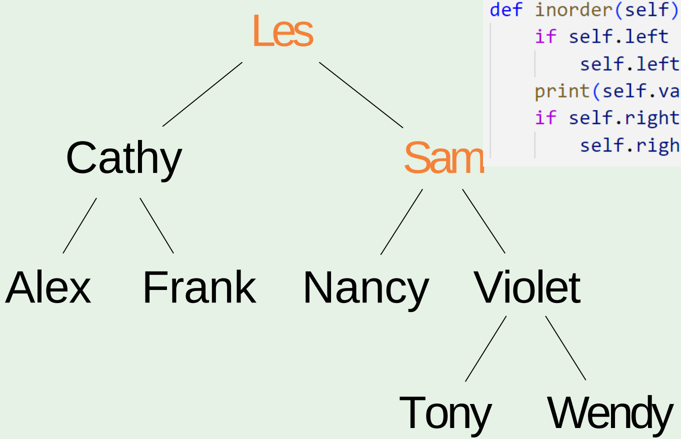


Output: Alex Cathy Frank Les Nancy Sam Tony Violet Wendy

InOrderTraversal




```python
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```

Output: Alex Cathy Frank Les Nancy Sam Tony Violet Wendy

InOrderTraversal

```python
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    if self.left is not None:
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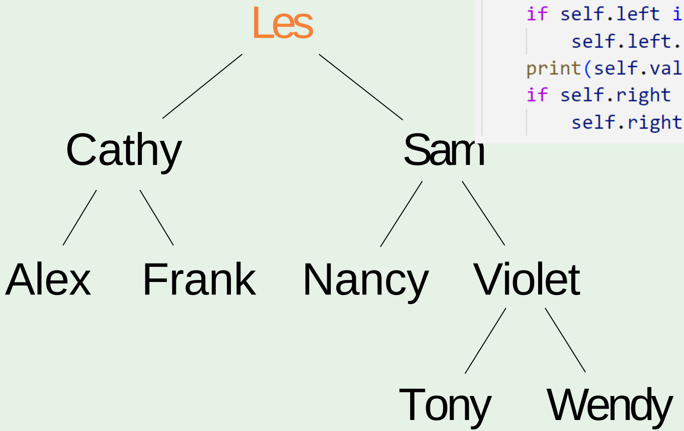

Output: Alex Cathy Frank Les Nancy Sam Tony Violet Wendy

InOrderTraversal

```python
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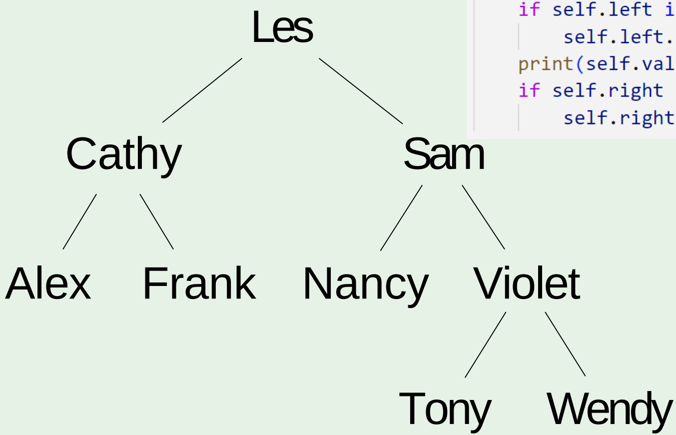


Output: Alex Cathy Frank Les Nancy Sam Tony Violet Wendy

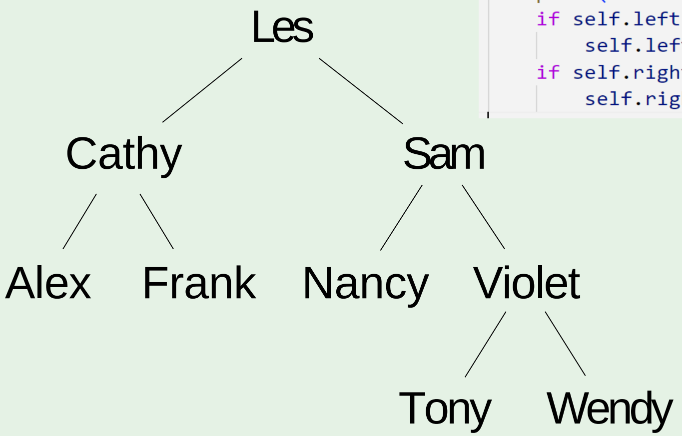
Depth-first

Pre-order traversal is to visit the root first. Then traverse the left subtree. Finally, traverse the right subtree.

```
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```

PreOrderTraversal

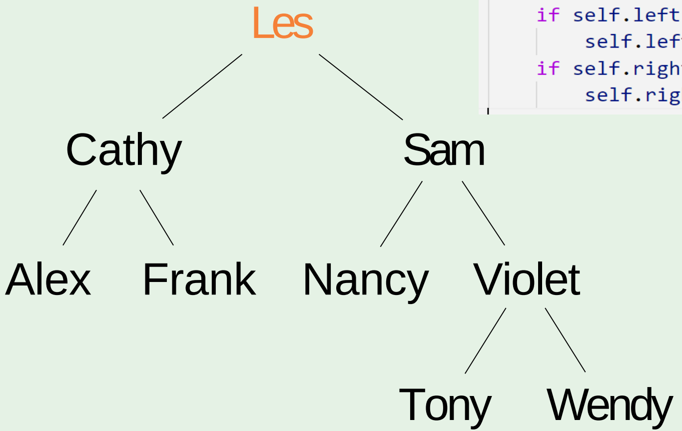
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output:

PreOrderTraversal

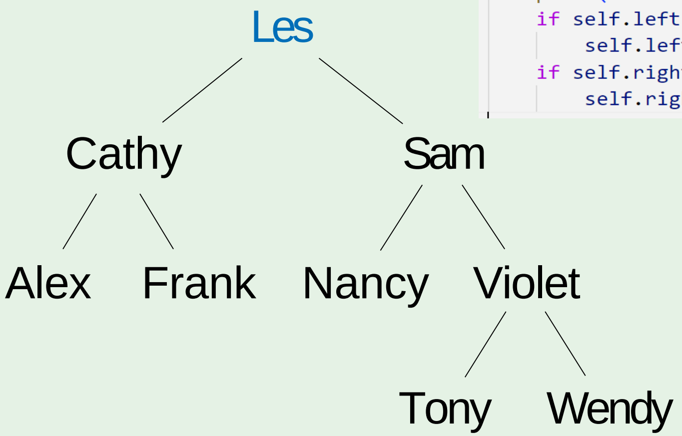
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output:

PreOrderTraversal

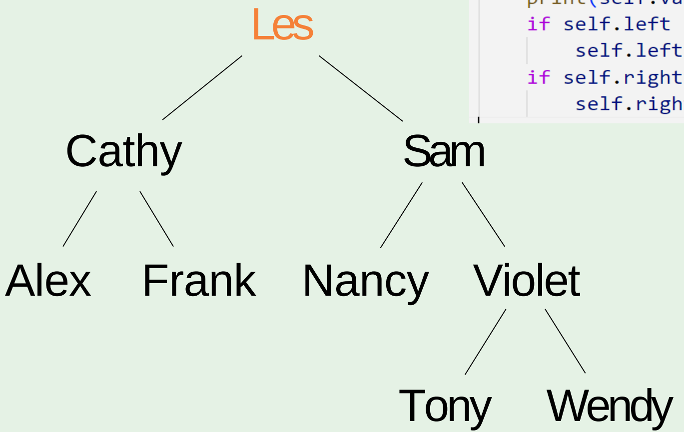
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les

PreOrderTraversal

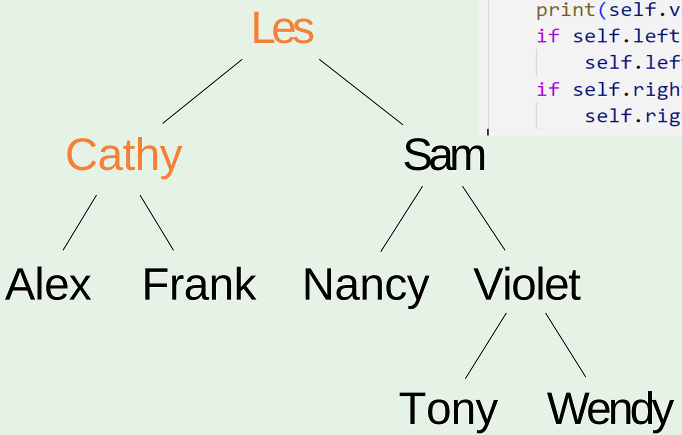
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les

PreOrderTraversal

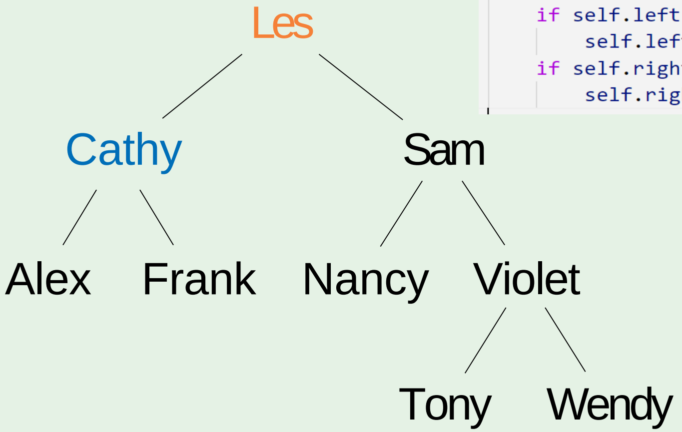
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les

PreOrderTraversal

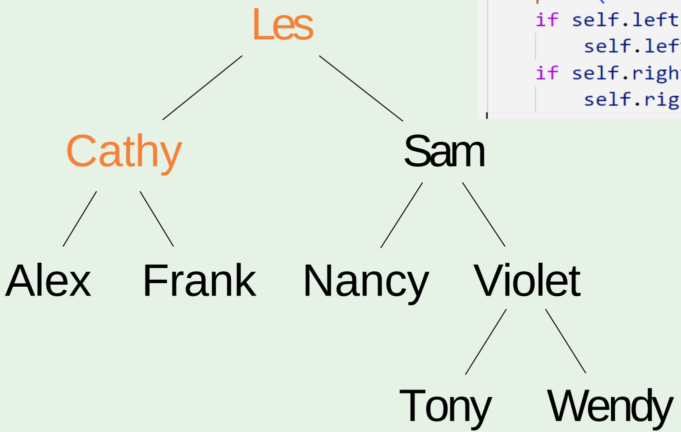
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy

PreOrderTraversal

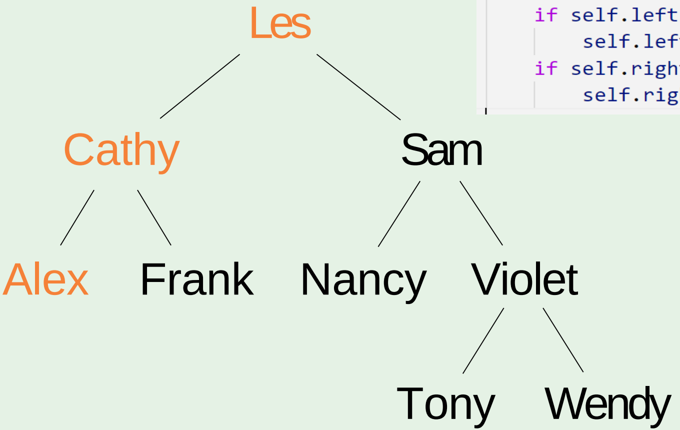
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy

PreOrderTraversal

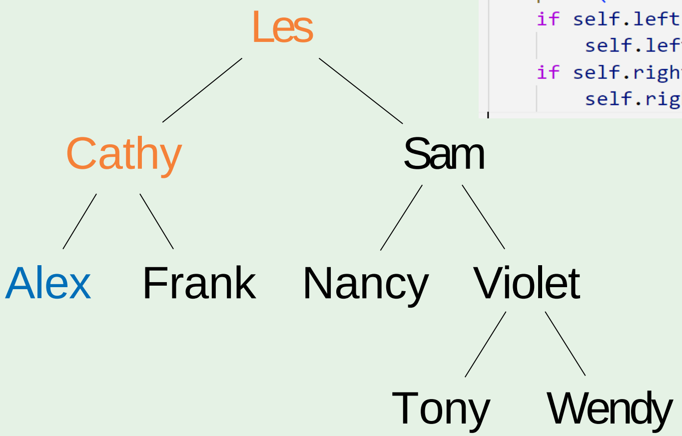
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy

PreOrderTraversal

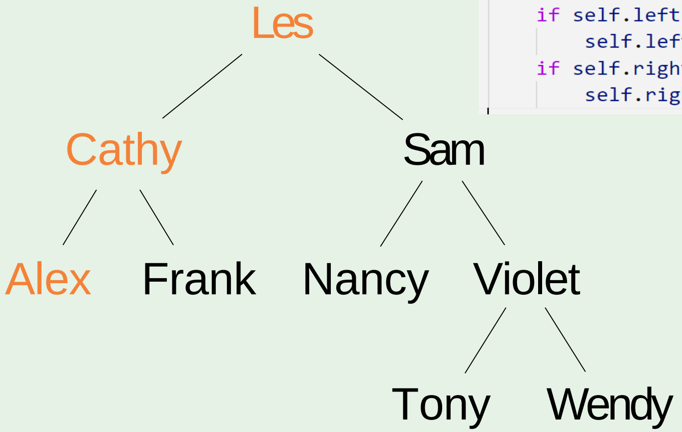
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex

PreOrderTraversal

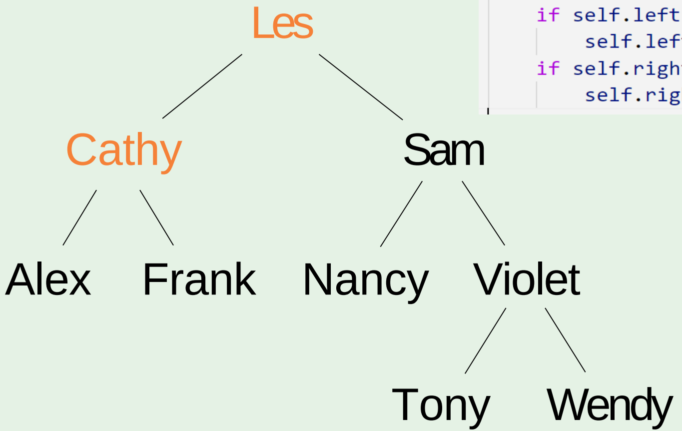
```
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex

PreOrderTraversal

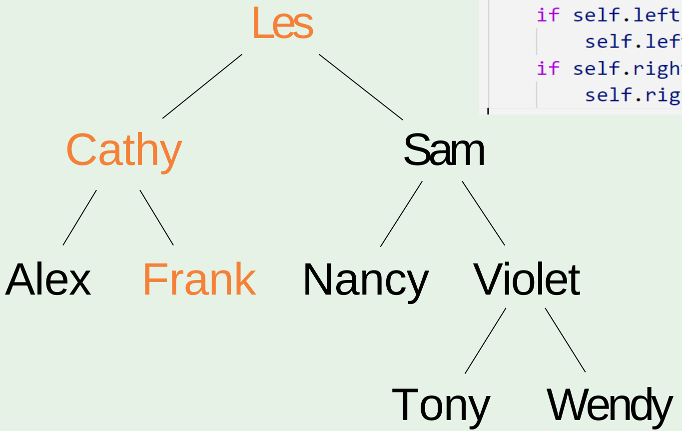
```
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex

PreOrderTraversal

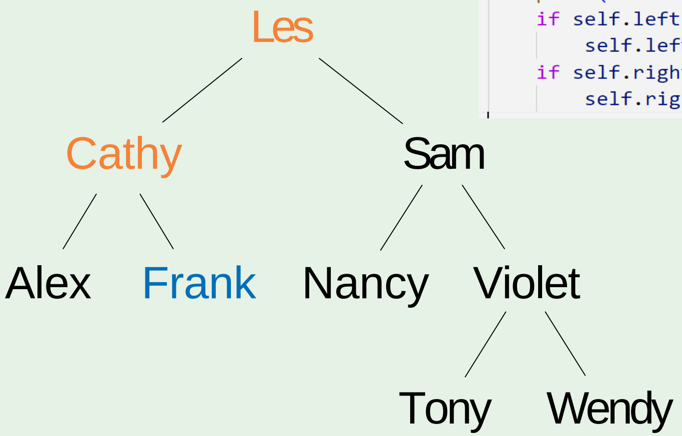
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex

PreOrderTraversal

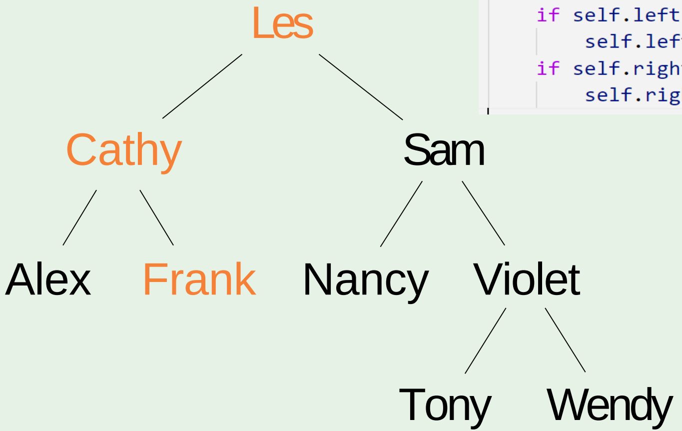
```
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex Frank

PreOrderTraversal

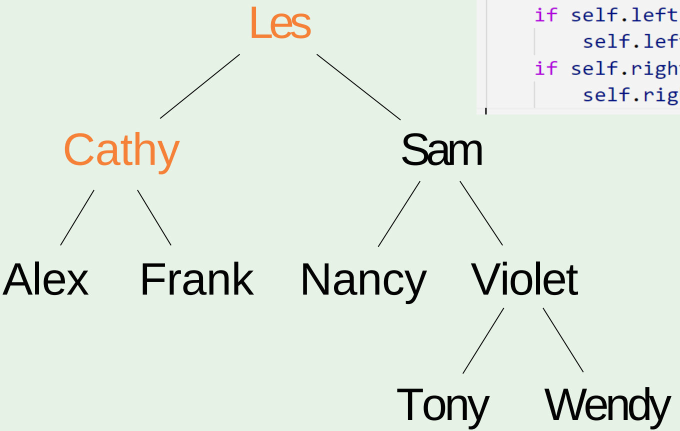
```
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex Frank

PreOrderTraversal

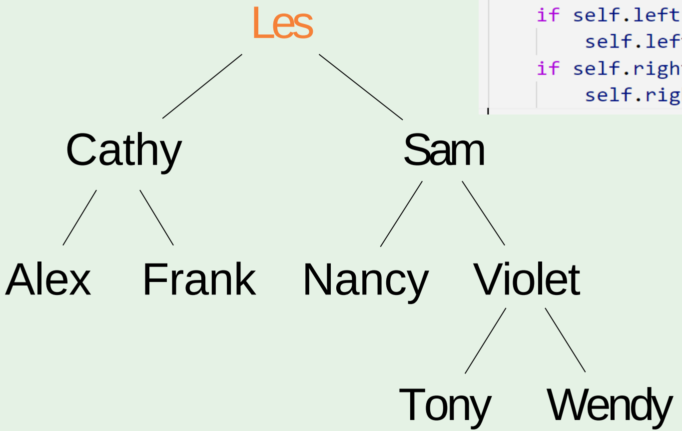
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex Frank

PreOrderTraversal

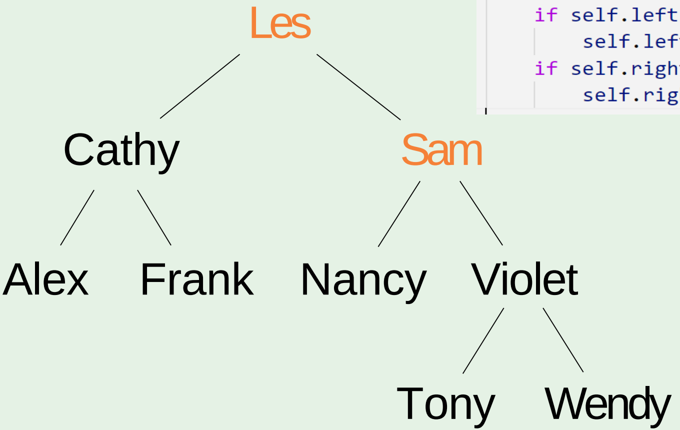
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex Frank

PreOrderTraversal

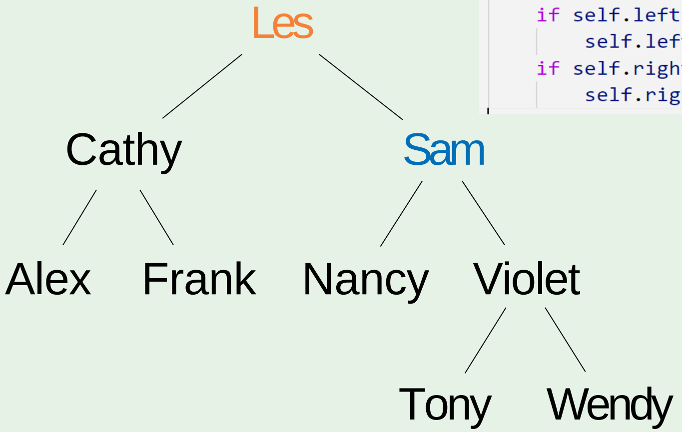
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex Frank

PreOrderTraversal

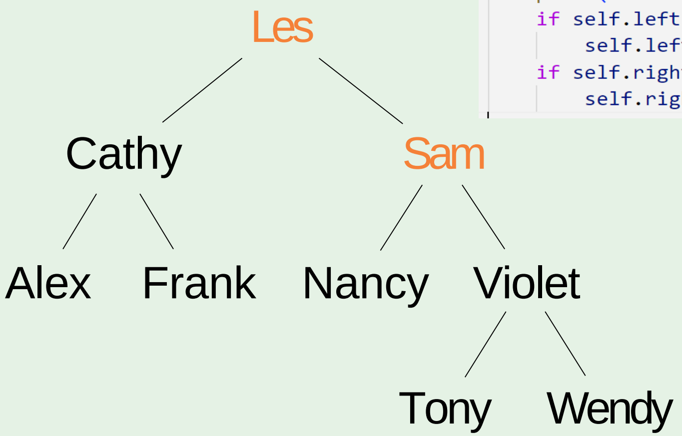
```
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex Frank Sam

PreOrderTraversal

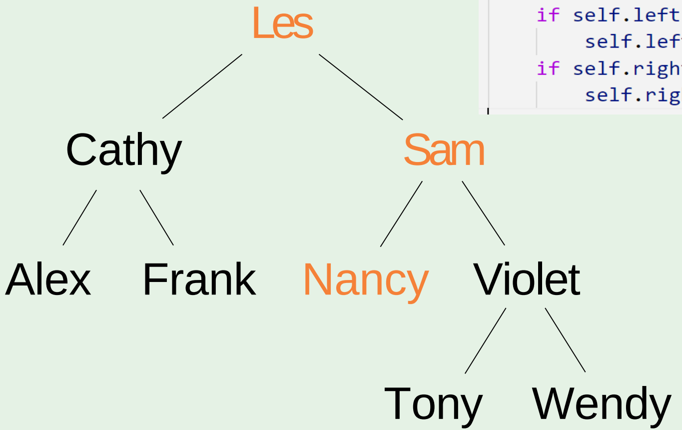
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex Frank Sam

PreOrderTraversal

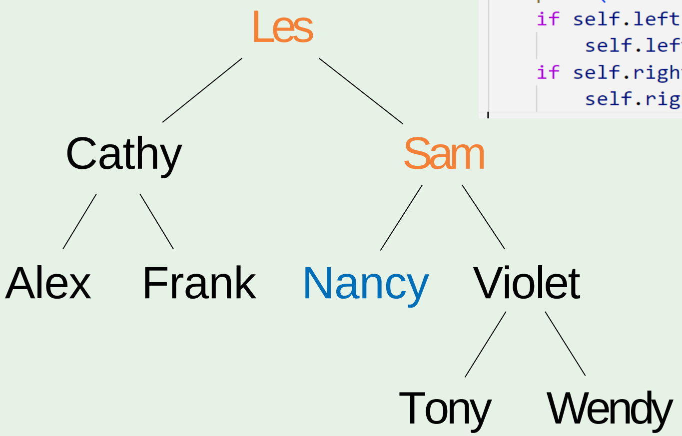
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex Frank Sam

PreOrderTraversal

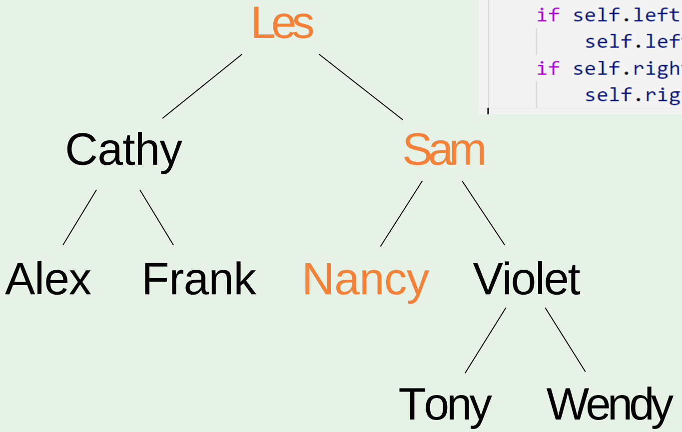
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex Frank Sam Nancy

PreOrderTraversal

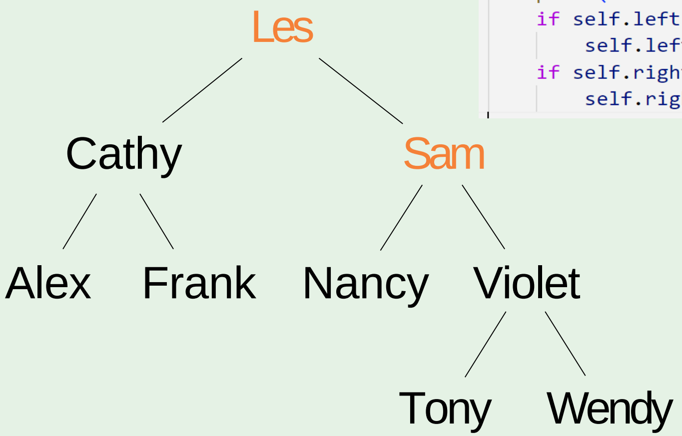
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex Frank Sam Nancy

PreOrderTraversal

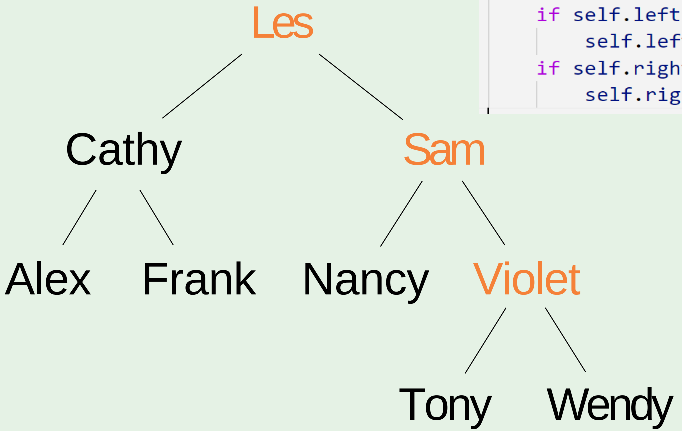
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex Frank Sam Nancy

PreOrderTraversal

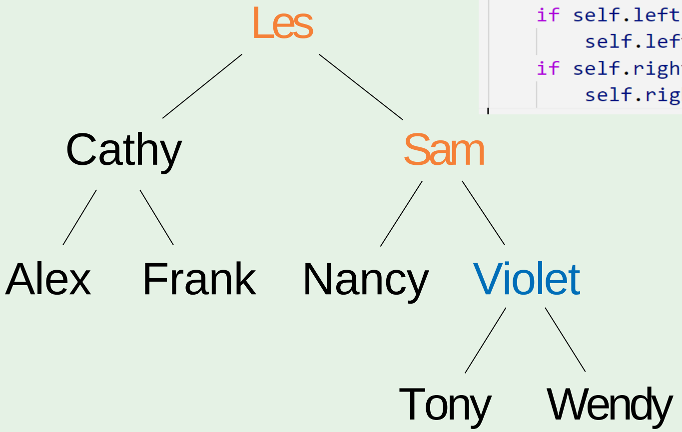
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex Frank Sam Nancy

PreOrderTraversal

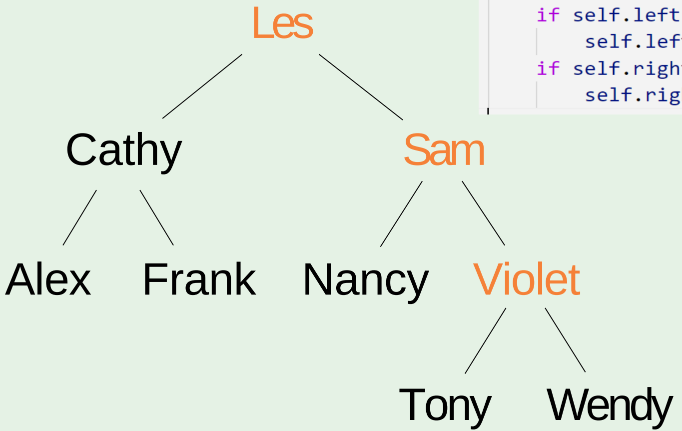
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex Frank Sam Nancy Violet

PreOrderTraversal

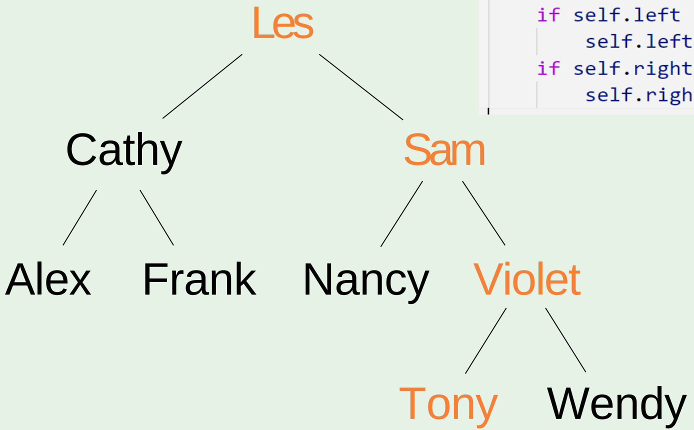
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex Frank Sam Nancy Violet

PreOrderTraversal

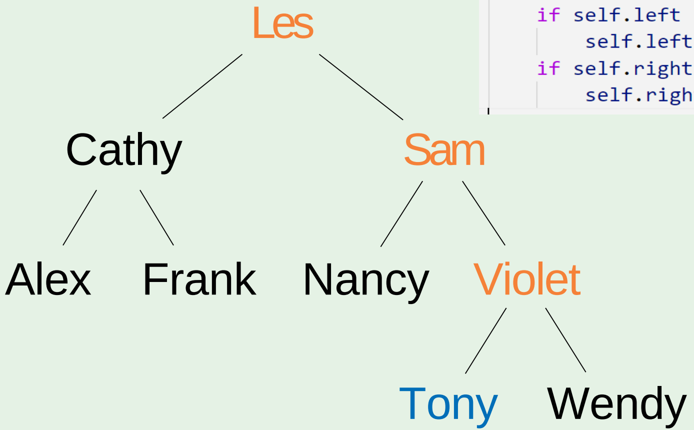
```
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex Frank Sam Nancy Violet

PreOrderTraversal

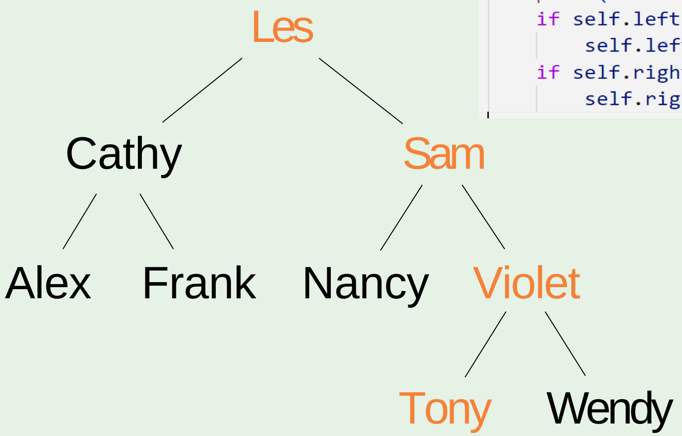
```
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex Frank Sam Nancy Violet Tony

PreOrderTraversal

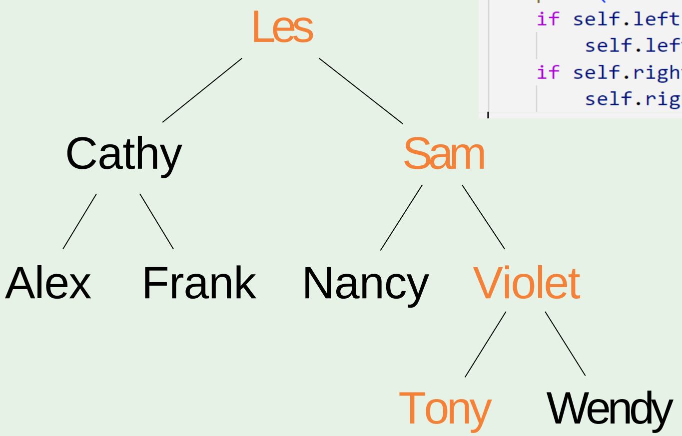
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex Frank Sam Nancy Violet Tony

PreOrderTraversal

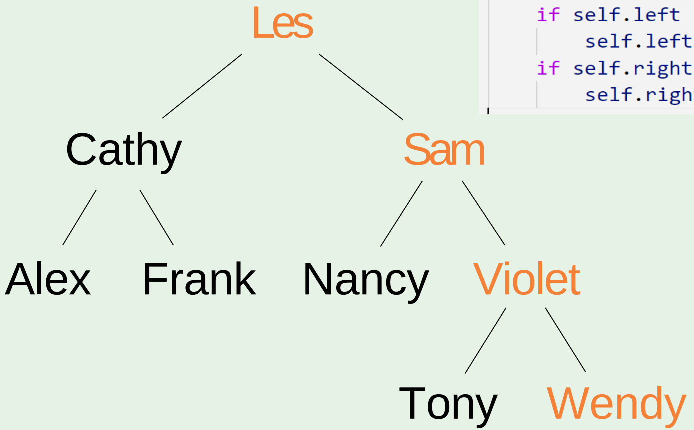
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex Frank Sam Nancy Violet Tony

PreOrderTraversal

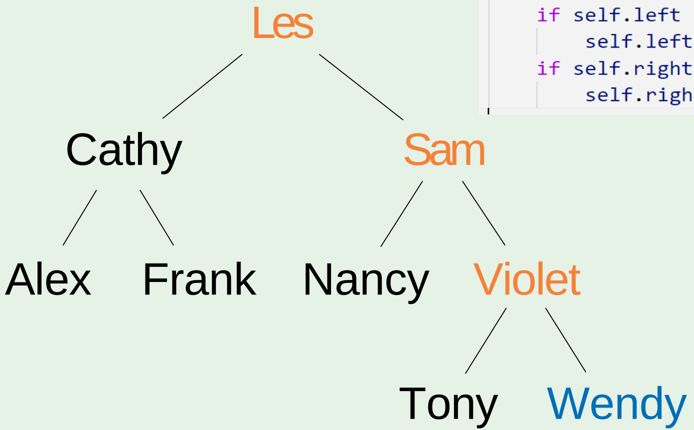
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex Frank Sam Nancy Violet Tony

PreOrderTraversal

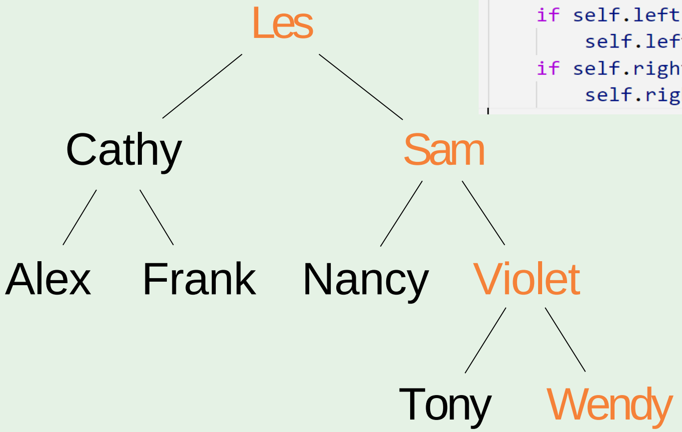
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex Frank Sam Nancy Violet Tony Wendy

PreOrderTraversal

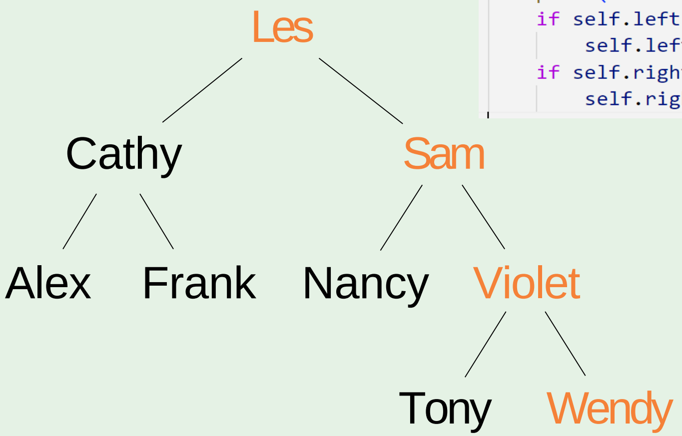
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex Frank Sam Nancy Violet Tony Wendy

PreOrderTraversal

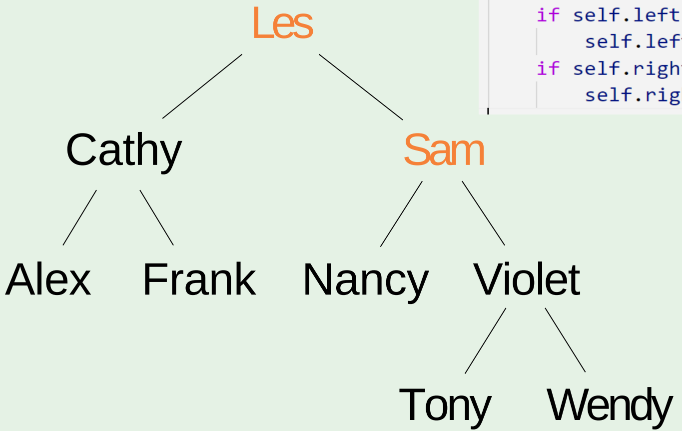
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex Frank Sam Nancy Violet Tony Wendy

PreOrderTraversal

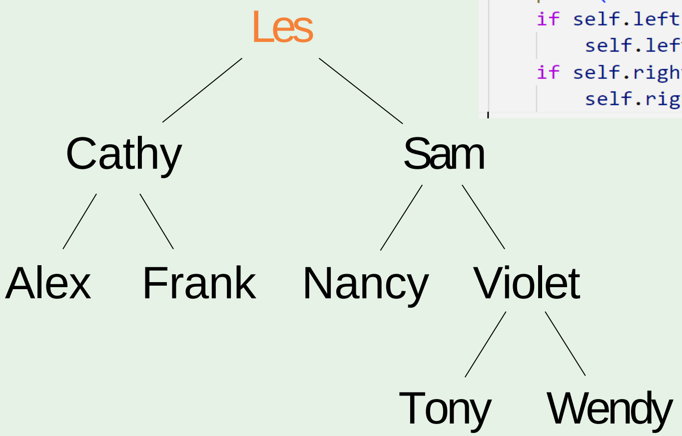
```
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex Frank Sam Nancy Violet Tony Wendy

PreOrderTraversal

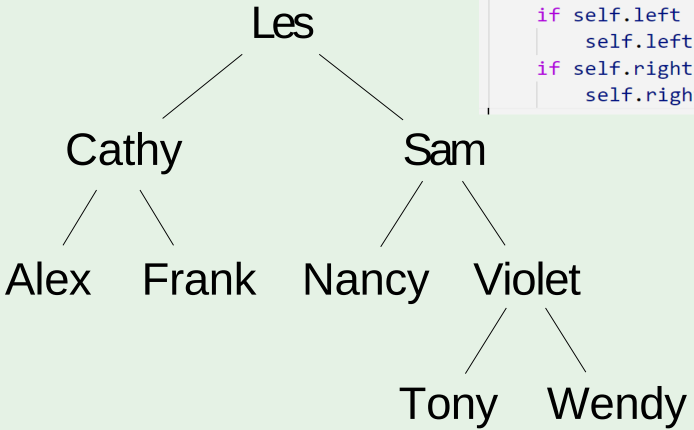
```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex Frank Sam Nancy Violet Tony Wendy

PreOrderTraversal

```python
def preorder(self):
    print(self.val)
    if self.left is not None:
        self.left.preorder()
    if self.right is not None:
        self.right.preorder()
```



Output: Les Cathy Alex Frank Sam Nancy Violet Tony Wendy

Depth-first

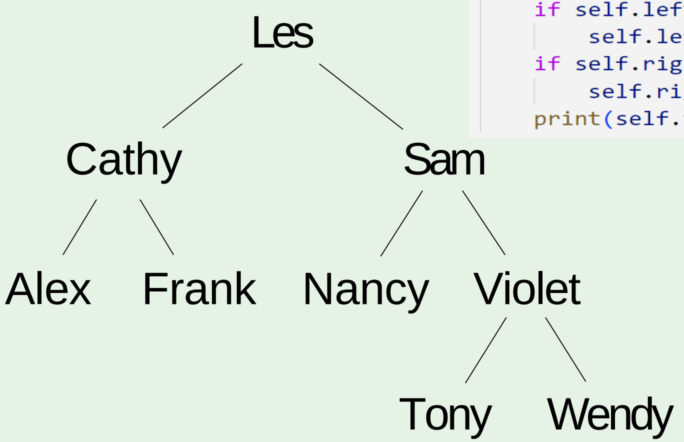
Post-order traversal is to traverse the left subtree first.

Then traverse the right subtree. Finally, visit the root.

```
def postorder(self):
    if self.left is not None:
        self.left.postorder()
    if self.right is not None:
        self.right.postorder()
    print(self.val)
```

PostOrderTraversal

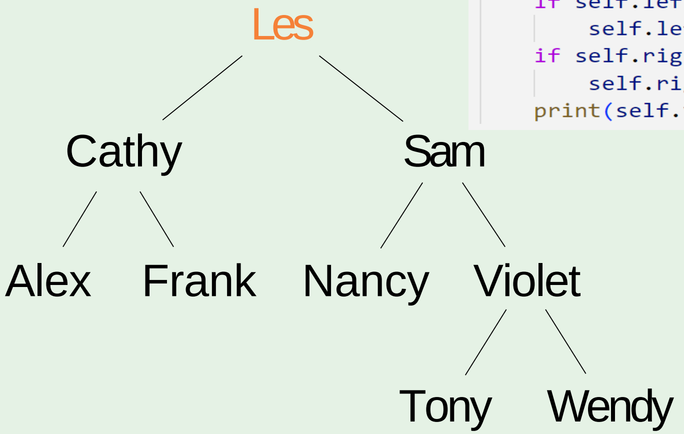
```python
def postorder(self):
    if self.left is not None:
        self.left.postorder()
    if self.right is not None:
        self.right.postorder()
    print(self.val)
```



Output:

PostOrderTraversal

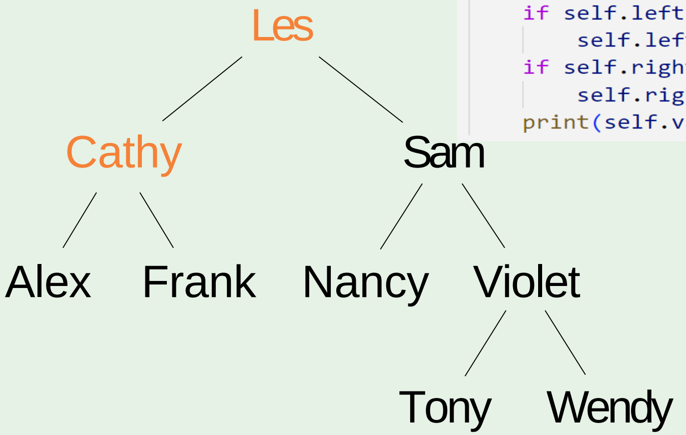
```
def postorder(self):
    if self.left is not None:
        self.left.postorder()
    if self.right is not None:
        self.right.postorder()
    print(self.val)
```



Output:

PostOrderTraversal

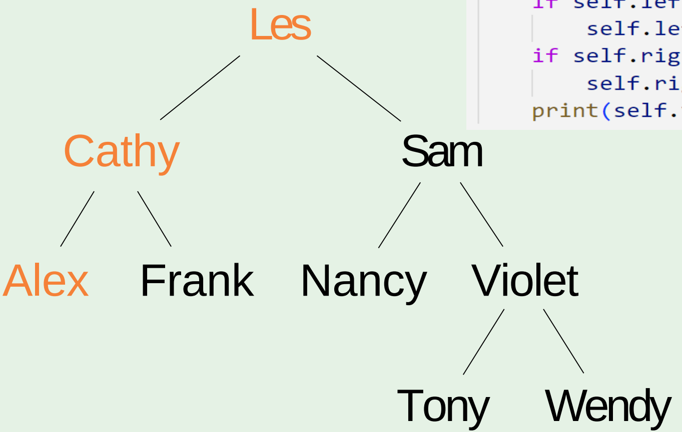
```python
def postorder(self):
    if self.left is not None:
        self.left.postorder()
    if self.right is not None:
        self.right.postorder()
    print(self.val)
```



Output:

PostOrderTraversal

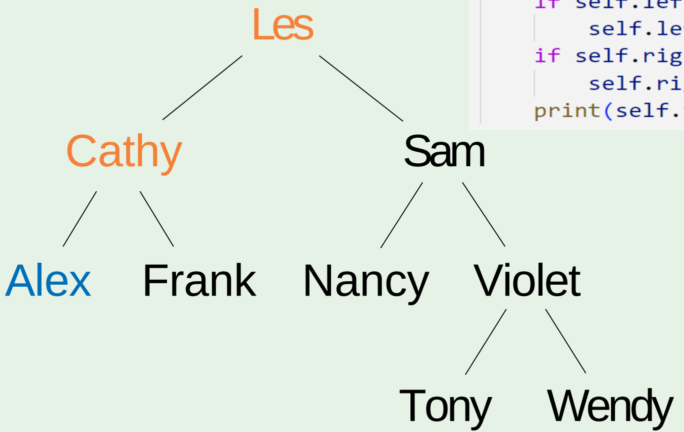
```
def postorder(self):
    if self.left is not None:
        self.left.postorder()
    if self.right is not None:
        self.right.postorder()
    print(self.val)
```



Output:

PostOrderTraversal

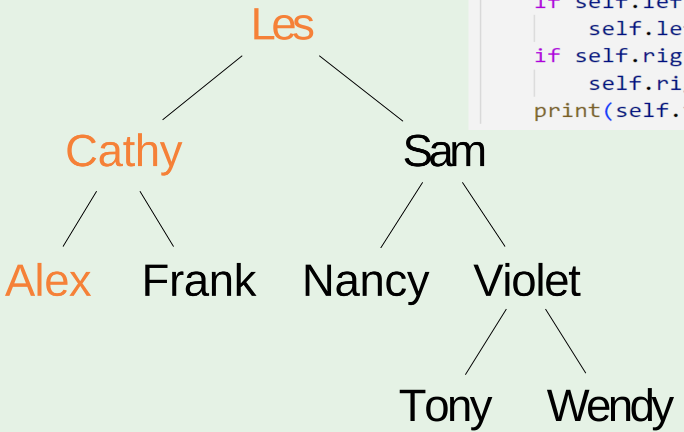
```
def postorder(self):
    if self.left is not None:
        self.left.postorder()
    if self.right is not None:
        self.right.postorder()
    print(self.val)
```



Output: Alex

PostOrderTraversal

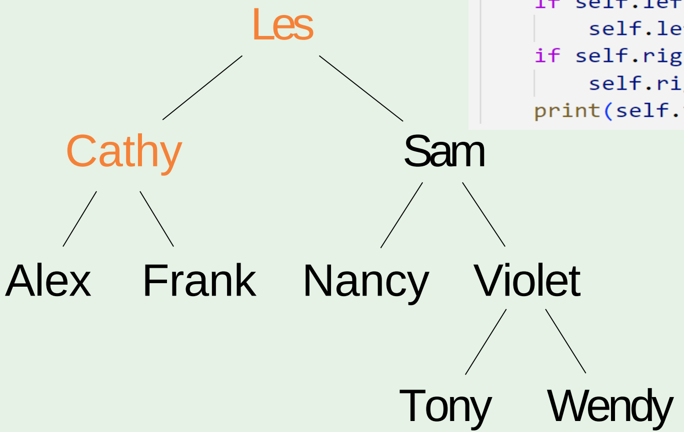
```python
def postorder(self):
    if self.left is not None:
        self.left.postorder()
    if self.right is not None:
        self.right.postorder()
    print(self.val)
```



Output: Alex

PostOrderTraversal

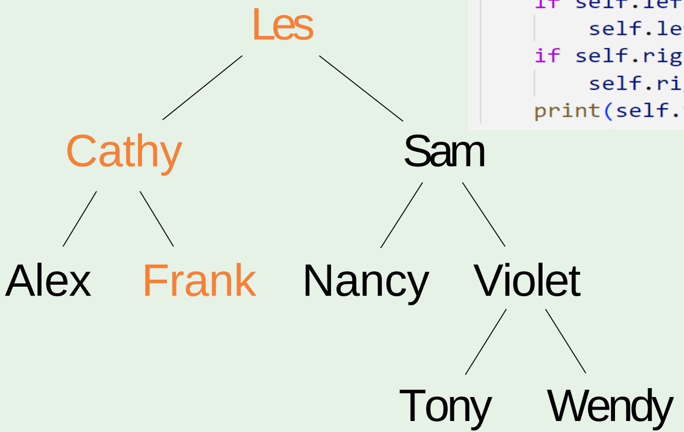
```python
def postorder(self):
    if self.left is not None:
        self.left.postorder()
    if self.right is not None:
        self.right.postorder()
    print(self.val)
```



Output: Alex

PostOrderTraversal

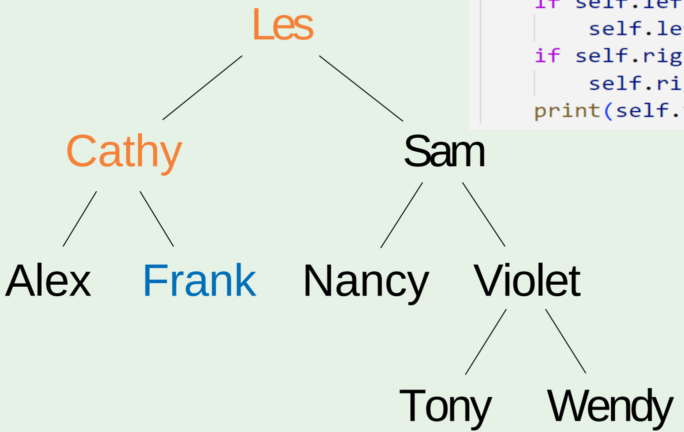
```python
def postorder(self):
    if self.left is not None:
        self.left.postorder()
    if self.right is not None:
        self.right.postorder()
    print(self.val)
```



Output: Alex

PostOrderTraversal

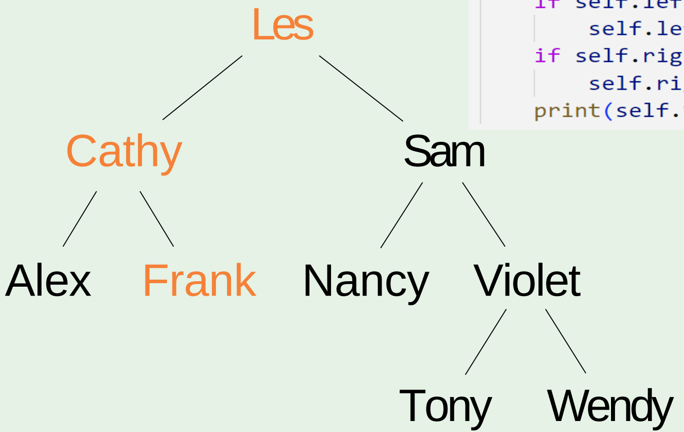
```python
def postorder(self):
    if self.left is not None:
        self.left.postorder()
    if self.right is not None:
        self.right.postorder()
    print(self.val)
```



Output: Alex Frank

PostOrderTraversal

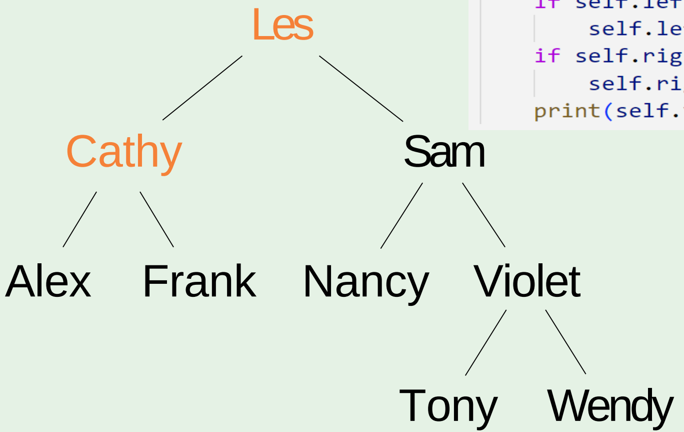
```
def postorder(self):
    if self.left is not None:
        self.left.postorder()
    if self.right is not None:
        self.right.postorder()
    print(self.val)
```



Output: Alex Frank

PostOrderTraversal

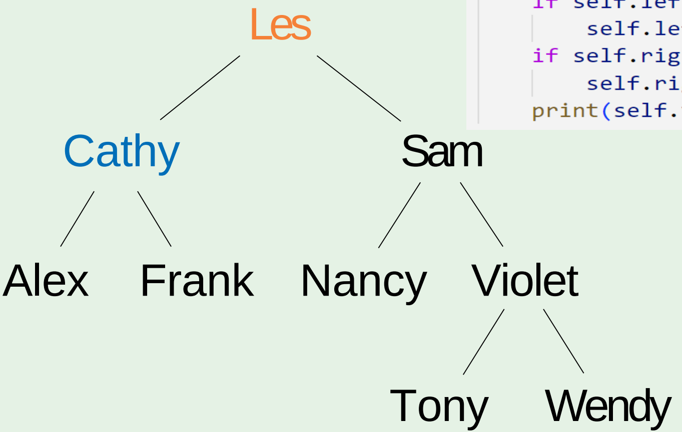
```python
def postorder(self):
    if self.left is not None:
        self.left.postorder()
    if self.right is not None:
        self.right.postorder()
    print(self.val)
```



Output: Alex Frank

PostOrderTraversal

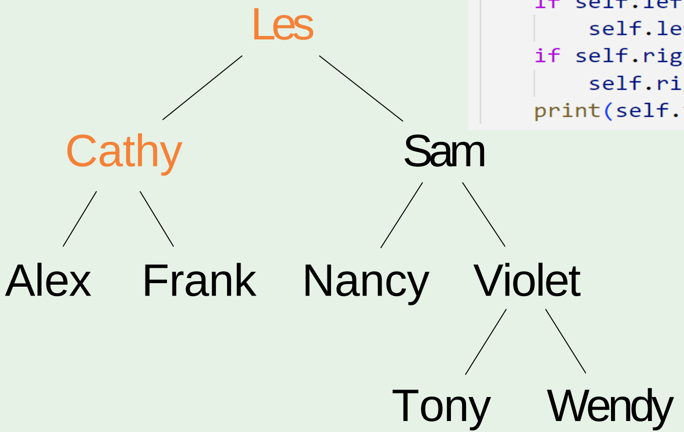
```python
def postorder(self):
    if self.left is not None:
        self.left.postorder()
    if self.right is not None:
        self.right.postorder()
    print(self.val)
```



Output: Alex Frank Cathy

PostOrderTraversal

```python
def postorder(self):
    if self.left is not None:
        self.left.postorder()
    if self.right is not None:
        self.right.postorder()
    print(self.val)
```



Output: Alex Frank Cathy

PostOrderTraversal

```python
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    if self.left is not None:
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    if self.right is not None:
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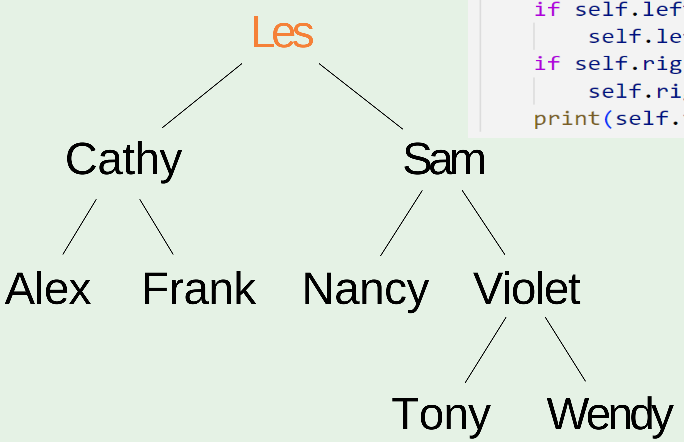

Output: Alex Frank Cathy

PostOrderTraversal

```python
def postorder(self):
    if self.left is not None:
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    if self.right is not None:
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    print(self.val)
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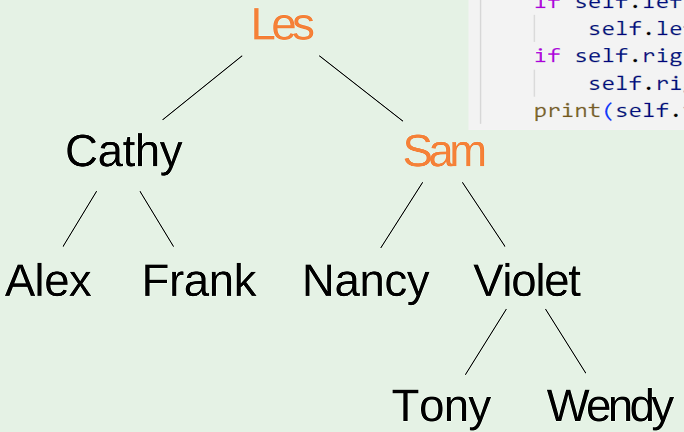

Output: Alex Frank Cathy

PostOrderTraversal

```
def postorder(self):
    if self.left is not None:
        self.left.postorder()
    if self.right is not None:
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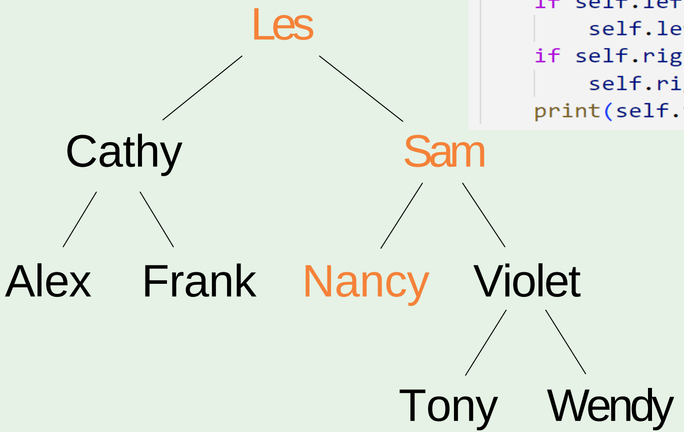

Output: Alex Frank Cathy

PostOrderTraversal

```python
def postorder(self):
    if self.left is not None:
        self.left.postorder()
    if self.right is not None:
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    print(self.val)
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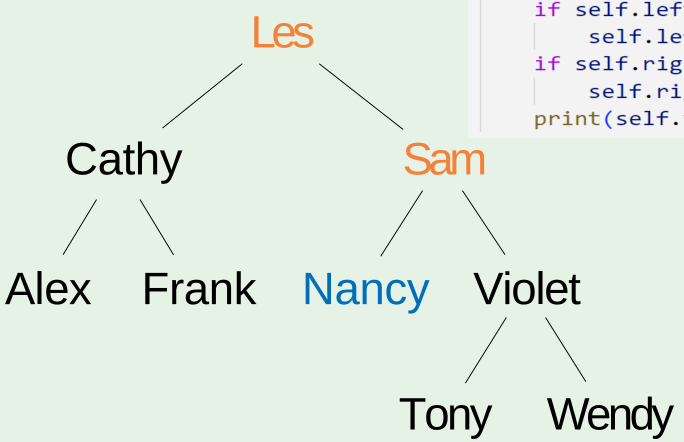
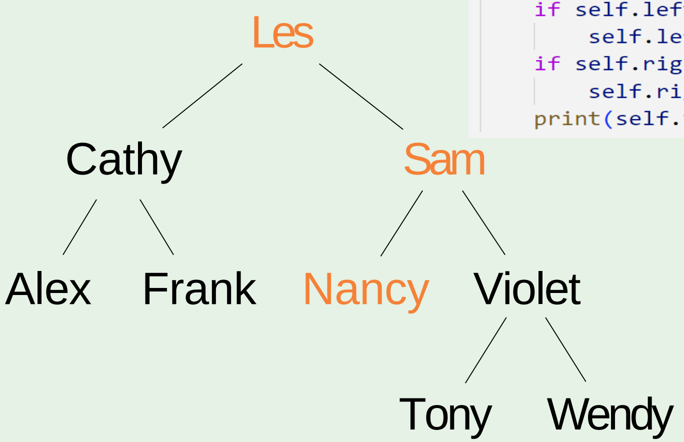


Output: Alex Frank Cathy Nancy

PostOrderTraversal

```
def postorder(self):
    if self.left is not None:
        self.left.postorder()
    if self.right is not None:
        self.right.postorder()
    print(self.val)
```



Output: Alex Frank Cathy Nancy

PostOrderTraversal

```python
def postorder(self):
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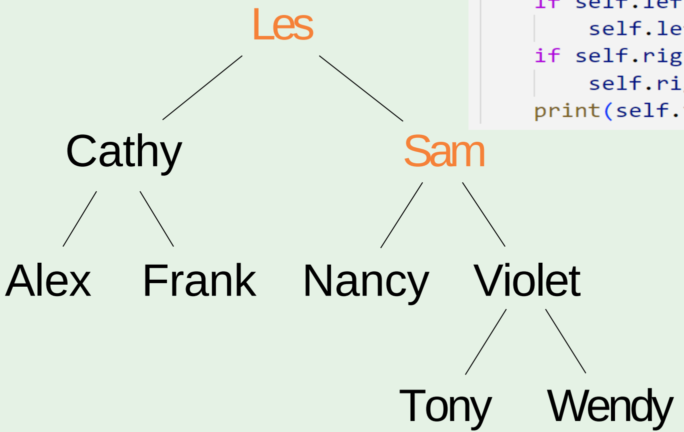

Output: Alex Frank Cathy Nancy

PostOrderTraversal

```python
def postorder(self):
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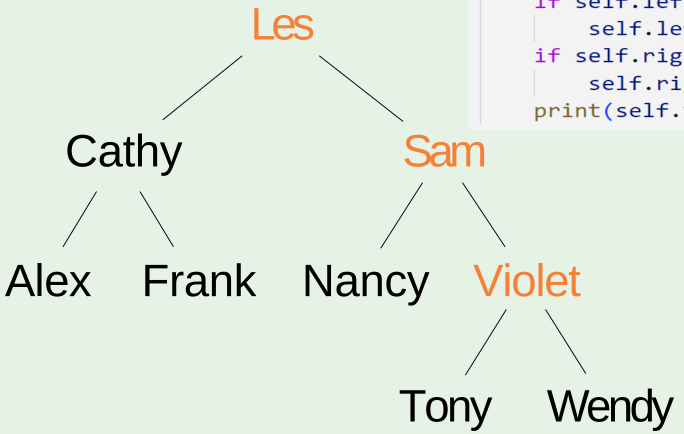

Output: Alex Frank Cathy Nancy

PostOrderTraversal

```python
def postorder(self):
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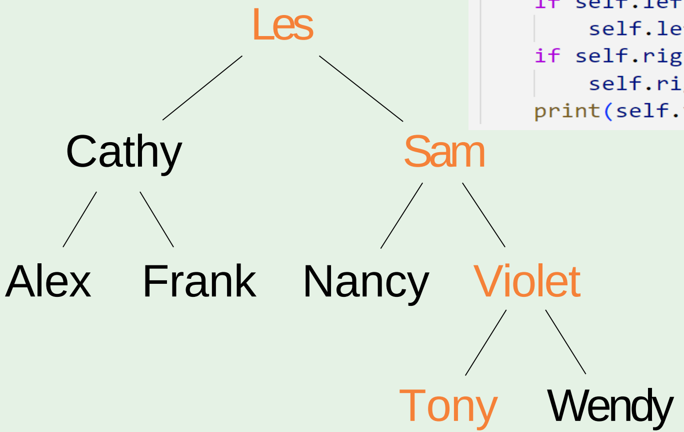

Output: Alex Frank Cathy Nancy

PostOrderTraversal

```python
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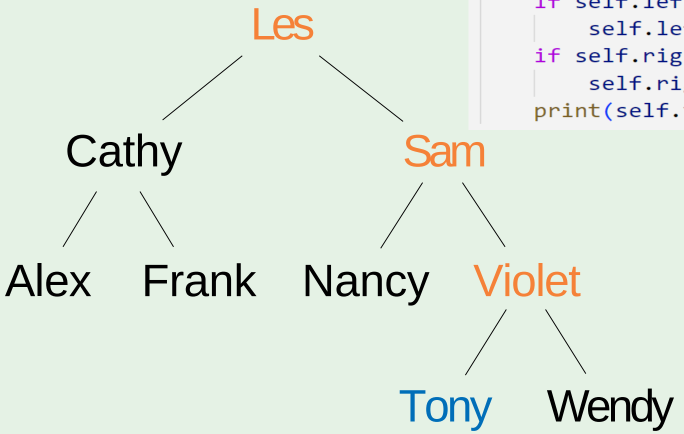

Output: Alex Frank Cathy Nancy Tony

PostOrderTraversal

```python
def postorder(self):
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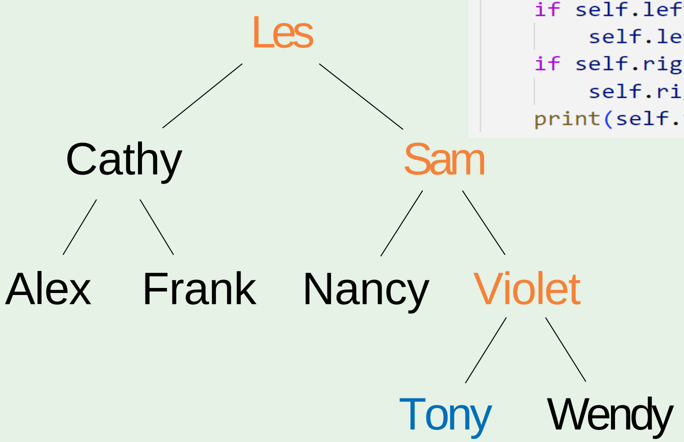

Output: Alex Frank Cathy Nancy Tony

PostOrderTraversal

```python
def postorder(self):
    if self.left is not None:
        self.left.postorder()
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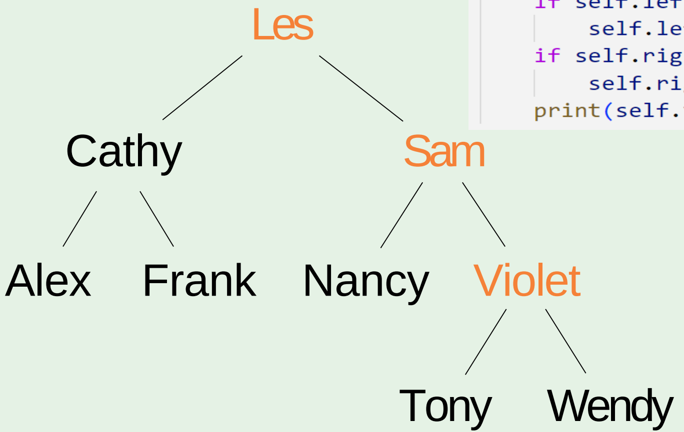

Output: Alex Frank Cathy Nancy Tony

PostOrderTraversal

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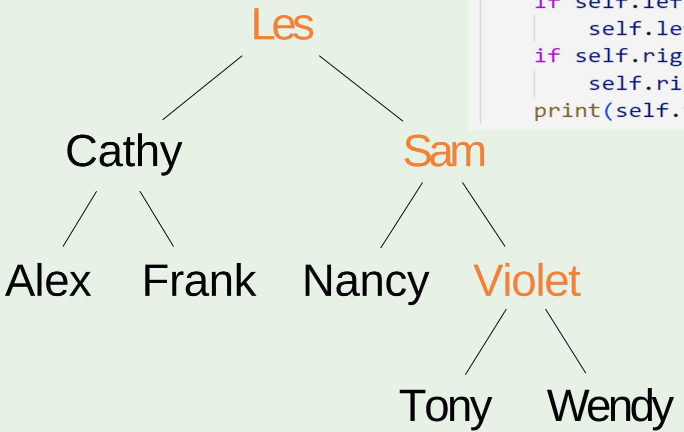

Output: Alex Frank Cathy Nancy Tony

PostOrderTraversal

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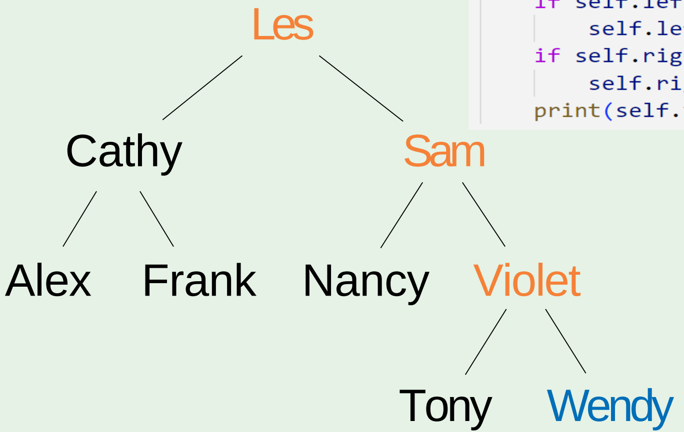

Output: Alex Frank Cathy Nancy Tony Wendy

PostOrderTraversal

```python
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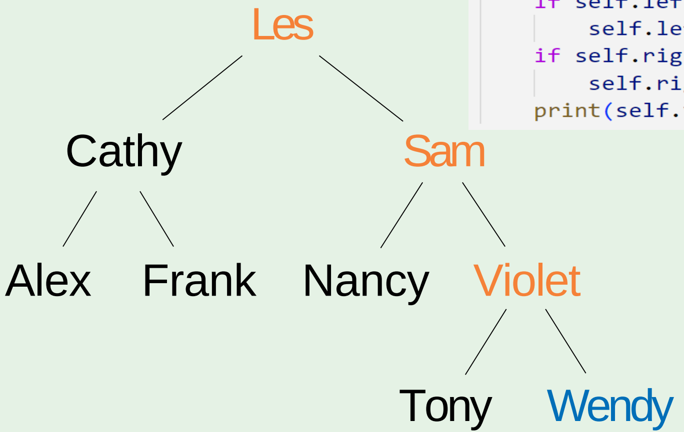

Output: Alex Frank Cathy Nancy Tony Wendy

PostOrderTraversal

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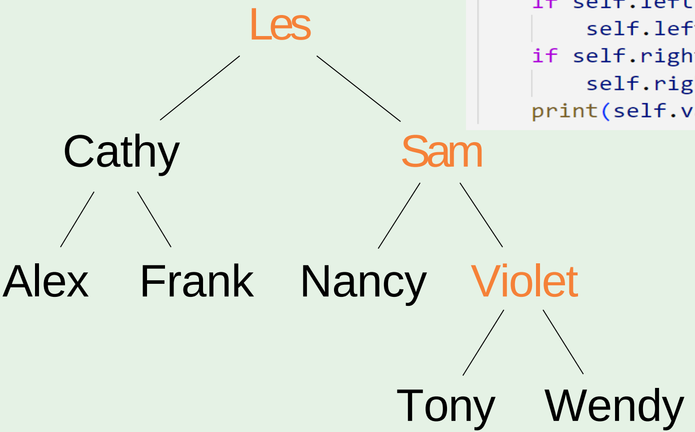

Output: Alex Frank Cathy Nancy Tony Wendy

PostOrderTraversal

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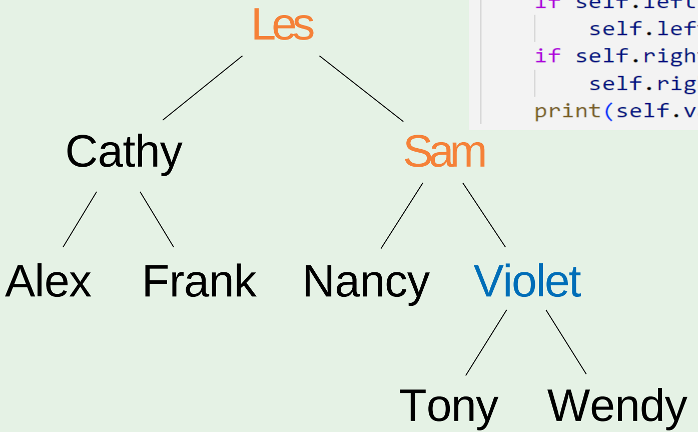

Output: Alex Frank Cathy Nancy Tony Wendy Violet

PostOrderTraversal

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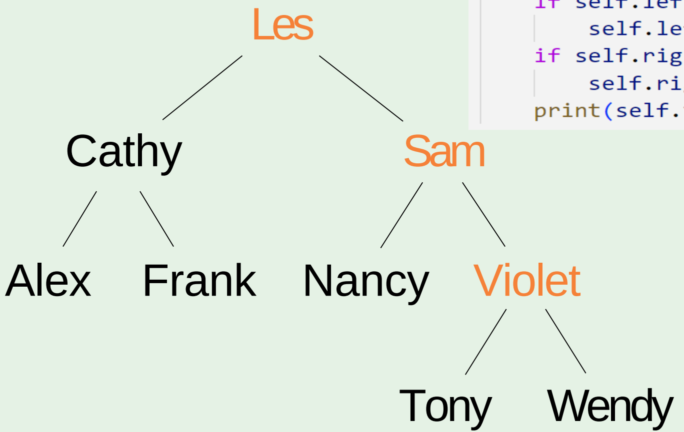

Output: Alex Frank Cathy Nancy Tony Wendy Violet

PostOrderTraversal

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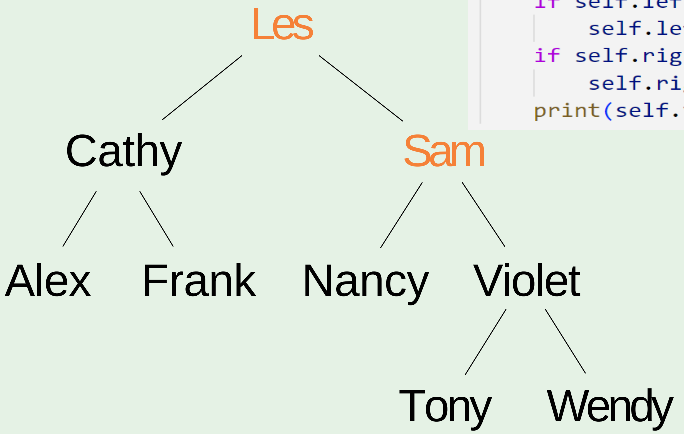

Output: Alex Frank Cathy Nancy Tony Wendy Violet

PostOrderTraversal

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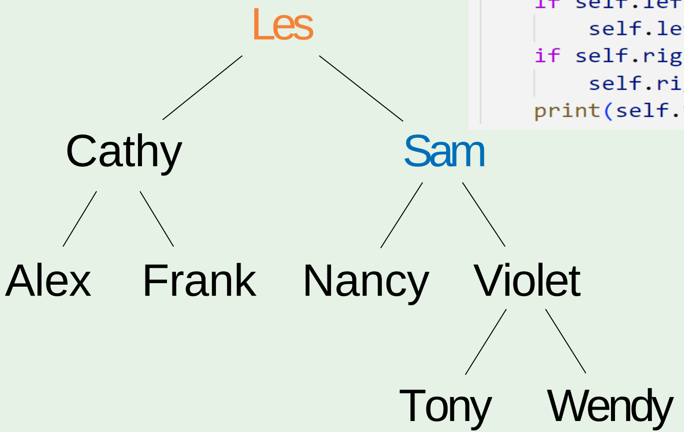

Output: Alex Frank Cathy Nancy Tony Wendy Violet Sam

PostOrderTraversal

```python
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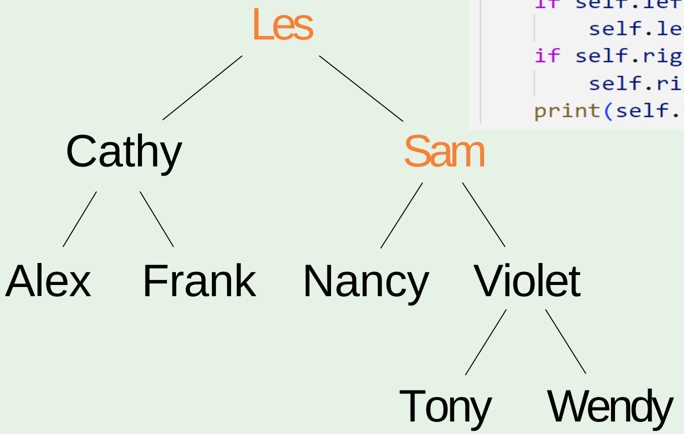

Output: Alex Frank Cathy Nancy Tony Wendy Violet Sam

PostOrderTraversal

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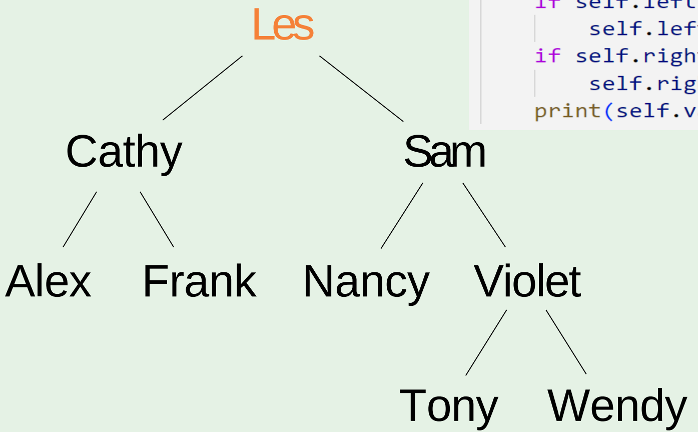

Output: Alex Frank Cathy Nancy Tony Wendy Violet Sam

PostOrderTraversal

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    if self.left is not None:
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    if self.right is not None:
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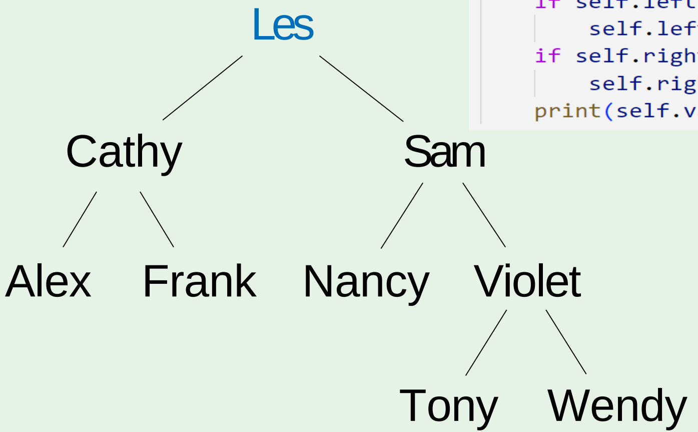

Output: Alex Frank Cathy Nancy Tony Wendy Violet Sam Les

PostOrderTraversal

```python
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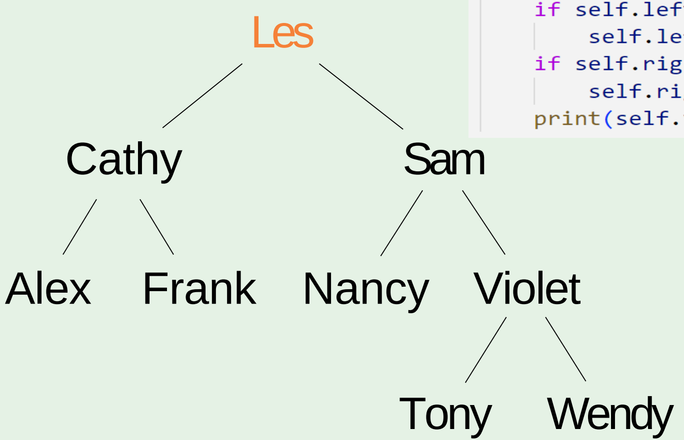

Output: Alex Frank Cathy Nancy Tony Wendy Violet Sam Les

PostOrderTraversal

```python
def postorder(self):
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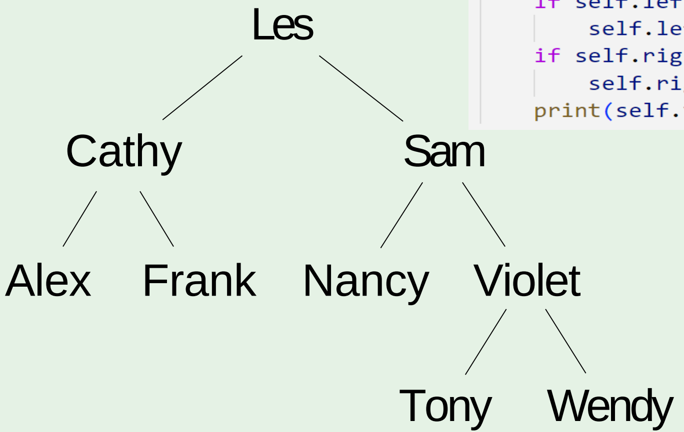


Output: Alex Frank Cathy Nancy Tony Wendy Violet Sam Les

Question

Assume: visiting a node is printing its label

Preorder:

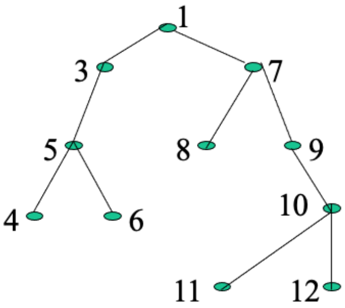
1 3 5 4 6 7 8 9 10 11 12

Inorder:

4 5 6 3 1 8 7 9 11 10 12

Postorder:

4 6 5 3 8 11 12 10 9 7 1



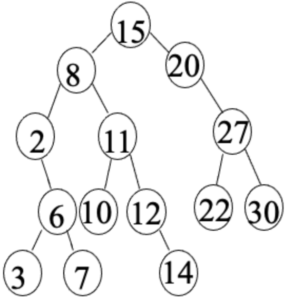
Question

Assume: visiting a node is printing its <u>data</u>

Preorder: 15 8 2 6 3 7 11 10 12 14 20 27 22 30

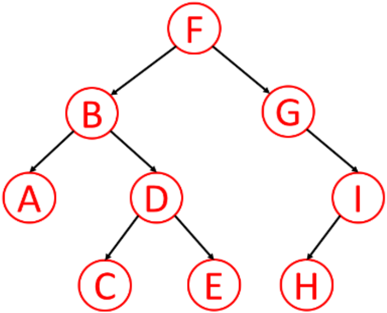
Inorder: 2 3 6 7 8 10 11 12 14 15 20 22 27 30

Postorder: 3 7 6 2 10 14 12 11 8 22 30 27 20 15



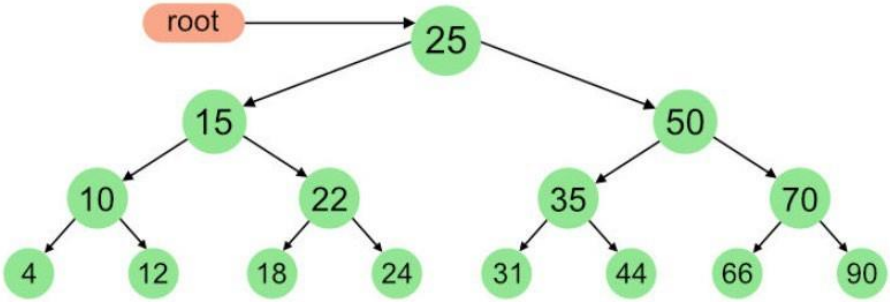
Question 1

For the tree below, write the in-order traversal.



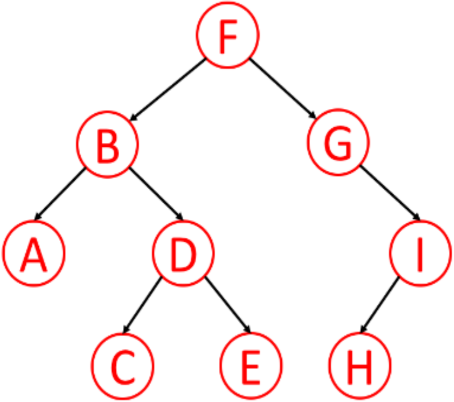
Question 2

For the tree below, write the pre-order traversal.



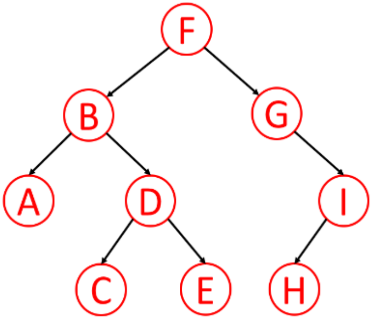
Question 3

For the tree below, write the pre-order traversal.



Question 4

For the tree below, write the post-order traversal.



Question 5

For the tree below, write the post-order traversal.